University of Portland SHILEY SCHOOL OF ENGINEERING

Outfall Basin 65 Stormwater Treatment

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Executive Summary

The Outfall Basin 65 Stormwater Treatment Project is a year-long senior capstone project in collaboration with the Portland Bureau of Environmental Services (BES) with the objective of treating stormwater runoff and reducing associated pollutant loading from the Outfall 65 basin into the Columbia Slough, which serves as a critical habitat for several wildlife species like fish and birds.

Simultaneously, we seek to address the urban heat island of the surrounding economically vulnerable neighborhoods. The first half of the project was focused on research and conceptual design. We identified various design considerations, conducted a site visit, and carried out extensive research on different treatment facility approaches. We also developed a stormwater model using EPA SWMM, through which we obtained flow values (results were lower than BES results) to further guide our conceptual design process. Five designs were established, and the final proposed design, consisting of two vaults on NE 11th Ave and NE 13th Ave and a green street along NE 11th, was chosen based on an design matrix outlining water quality performance, cost, environmental and social equity, and constructability. In the second half of the project, we will develop the design to 60% completion.

Introduction

The senior capstone project is a culminating experience in which students pursue a real-life project, gathering hands-on practice of engineering and invaluable professional skills. The Outfall 65 Stormwater treatment team is headed by Gus Shelley, Isabella Small, Maya Struzak, and Jayson Tagala and supervised by faculty, industry, and course advisors: Dr. Jordyn Wolfand, Daniel Jenkins (City of Portland Bureau of Environmental Services), and Dr. Sean Gestson.

The goal of this project is to design a stormwater treatment system that will treat runoff from the City Right of Way (ROW) draining to Outfall 65 (Figure 1). The desired outcome is to treat the stormwater runoff and reduce associated pollutant loading from the outfall basin.

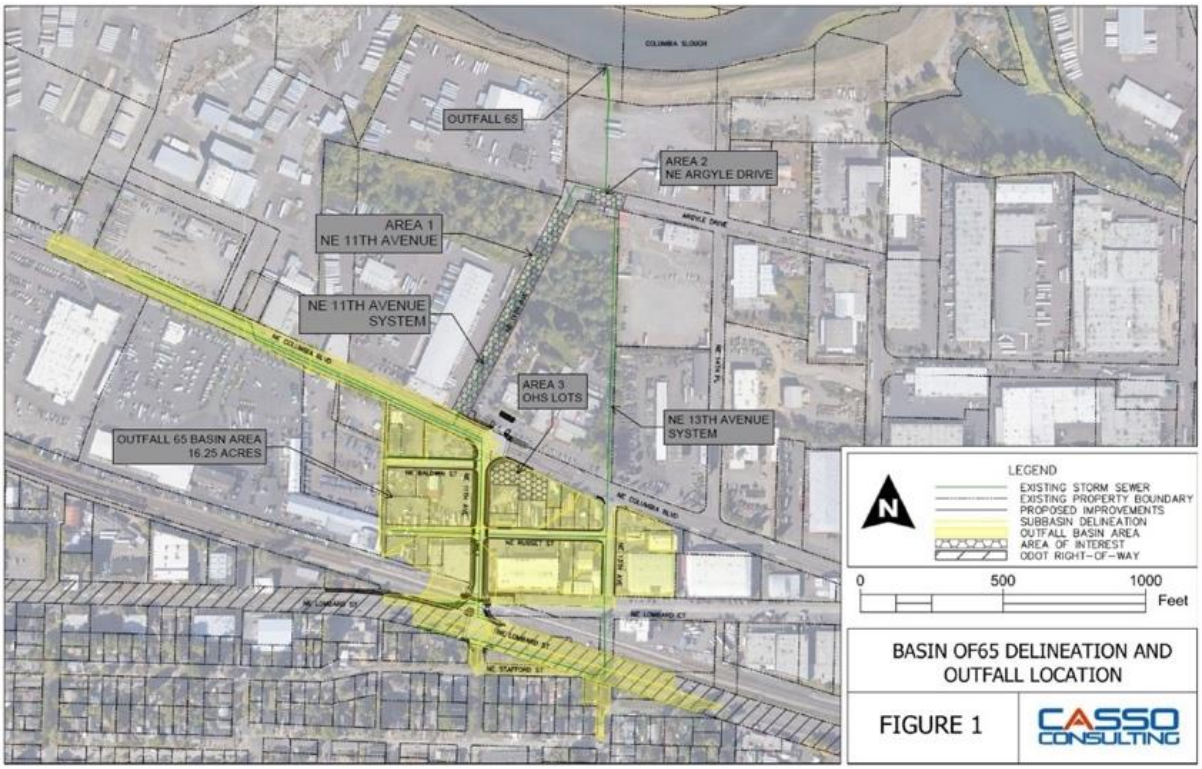

Figure 1. Site Map with basin delineation and outfall location.

Background

Geography

The Columbia River is the largest North American river flowing into the Pacific Ocean.[1] It covers 258,000 square miles in its 1,200-mile course to the ocean, flowing through parts of seven states, one Canadian province, and four mountain ranges.[2]

The Columbia Slough, in the floodplain of the Columbia River, is a waterway stretching about 20 miles, with a basin area of approximately 51 square miles. Beginning in Fairview, Oregon it moves west through Portland into the Willamette River. Outfall 65, one of the 850 outfalls along the Slough, is in North Portland and captures untreated stormwater over an area of about 16 acres.

History

The Columbia Slough is home to the Upper Chinook native people, resting on traditional village sites of the Multnomah, Clackamas, and Cascade bands of Chinook and the Tualatin band of Kalapuya. The Tribes used the Slough and the surrounding areas for hunting, fishing, and travel. Through sophisticated land stewardship systems, they were able to "protect rich and productive waterways, wetlands, and uplands".[3] However, European settlement in the early 19th century led to the forced removal of Native people and urban development of cities such as Portland.[4]

To accommodate industry, agriculture, and transportation, the watersheds were altered beginning in 1918 for flood control and land development. Levee construction formed three drainage districts, and wetlands, streams, and side channels were filled and diverted. These alterations led to a major loss of natural habitats such as wetlands, floodplains, and forests, transforming the slough into a highly managed and channelized system.

Habitats

Despite the major alterations and habitat loss, the Columbia Slough serves as an important wildlife corridor, rich in biodiversity. It houses a wide range of species, many of which are found nowhere else in Portland. It is home to multiple mammal species such as beaver and deer, about 175 bird species, and 29 fish species. Additionally, it is a critical habitat for salmon, and steelhead trout use the slough when migrating to the ocean.[5]

Despite its ecological value, it currently faces multiple threats such as pollution and habitat loss and deterioration. A major source of pollution is from stormwater runoff. As runoff flows over impervious surfaces, it collects and transports contaminants into the Slough; this is exacerbated by the industrial nature of the surrounding sites. The main types of contaminants of concern are metals, pesticides, PAHs,

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[1]Marts, Marion E. "Columbia River." Encyclopædia Britannica, November 30, 2023. https://www.britannica.com/place/Columbia-River.

[2] "Columbia River." American Rivers, 2023. https://www.americanrivers.org/river/columbia-river/.

[3] "Columbia Slough Watershed." Portland.gov, 2023. https://www.portland.gov/bes/protecting-rivers-streams/portlands-watersheds/columbia-slough.

[4] "About the Watershed." Columbia Slough Watershed Council, 2023. https://www.columbiaslough.org/about-the-watershed.

[5] "Columbia Slough Watershed." Portland.gov, 2023. https://www.portland.gov/bes/protecting-rivers-streams/portlands-watersheds/columbia-slough.

and PCBs.[6] Some of these contaminants bioaccumulate in fish that live in the Slough, which are harmful to eat as a result. The City of Portland has a fish advisory for the Columbia Slough, which "recommends eating only one fillet meal each month for all fish caught in the slough". There are currently various efforts to amend these issues, including city projects to improve habitats, protect natural spaces, and improve water quality. [7]

Urbanization

Today, the area around the slough is home to about 170,000 people and supports 4,200 businesses. According to the City of Portland, many of the neighborhoods around the Slough are low-income and experience high economic vulnerability (Appendix H). Populated by many underserved communities, the areas surrounding the Slough can be low equity.

The 'Urban Heat Island' is a term referring to the temperature increase in urban and industrialized areas due to the absorption and reflection of sun rays on impervious surfaces.[8] The lack of vegetation and urbanization are contributing factors to Portland's urban heat island. Impervious surfaces, such as concrete pavement, trap heat and prevent cooling processes brought on by the natural water cycle. A recent NASA study on urban heat vulnerability in Portland shows that communities around the Columbia Slough are subject to high land surface heat and are considered highly vulnerable to the effects of climate change.[9]

See Appendix H for associated maps (land use, soil type, economic vulnerability, urban heat).

Stormwater Management

Gray infrastructure encompasses traditional stormwater infrastructure like drains, pipes, and basins. Despite the wide use of gray infrastructure, green infrastructure (GI) has emerged as a nature-based solution, mimicking the processes of the natural water cycle. By promoting infiltration and evapotranspiration, GI contributes to stormwater runoff reduction, cooling, aquifer recharge, and water quality improvement via passive treatment. In optimizing both gray and green infrastructure, communities can expect more resilient stormwater systems as well as environmental, social, and economic benefits.[10] The addition of green spaces and vegetation like trees, green roofs, and vegetation, can help reduce urban heat island effects by shading building surfaces, deflecting radiation from the sun, and releasing moisture into the atmosphere.

Stormwater runoff occurs when excess precipitation flows over impervious surfaces instead of infiltrating into the ground. Urbanization increases the volume and rate of stormwater runoff due to the removal of vegetation, compaction of soils, and impervious surfaces. Heavy metals and oils, among others, are expected to be present in runoff in industrial areas like the Outfall 65 basin area. These contaminants are

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[6] DEQ. "Environmental Cleanup Site Information (ECSI) Database Site Summary Report - Details for Site ID 1283, Columbia Slough." State of Oregon: Department of Environmental Quality, 2023. https://www.deq.state.or.us/lq/ECSI/ecsidetail.asp?seqnbr=1283.

[7] "Columbia Slough Watershed." Portland.gov, 2023. https://www.portland.gov/bes/protecting-rivers-streams/portlands-watersheds/columbia-slough.

[8] "Heat Island Effect." United States Environmental Protection Agency, 2023. https://www.epa.gov/heatislands.

[9] Kessler, Keegan. Tech. Portland Urban Development: Quantifying and Visualizing Urban Heat with Compounding Vulnerabilities to Support Community Depaving Initiatives. Develop, 2023.

[10] "What Is Green Infrastructure? ." United States Environmental Protection Agency, 2023. https://www.epa.gov/green-infrastructure/what-green-infrastructure.

picked up by stormwater and either create pooled surfaces or end up in the slough. The City of Portland Bureau of Environmental Services (BES) manages Portland's stormwater network, consistently planning, maintaining, or enhancing watershed systems. The Columbia Slough Sediment Program, led by BES and the Oregon Department of Environmental Quality (DEQ), was established to address pollution in the Slough by reducing contaminant discharges. According to the Portland Stormwater Management Manual, "City-required stormwater retrofits are required by the City as a result of City code, rule implementation, or an enforcement action. City-required stormwater retrofits must meet the [Stormwater Management Manual] requirements in full."

Managing the Outfall 65 basin is necessary because there are no existing treatment facilities. Stormwater management can be approached by modeling the Outfall 65 basin using the Environmental Protection Agency's Stormwater Management Model (SWMM). This model is widely used for planning, analysis, and design related to stormwater. With a developed model, various storms can be simulated, and the runoff can be quantified, which is crucial for the design, sizing, and placement of the proposed facilities.

Design Considerations and Constraints

Public Health, Safety, and Wellbeing

The Columbia Slough and Portland communities can be greatly enhanced by treating stormwater pollutants and mitigating the risk of flooding. Treating stormwater will improve the water quality for marine life, restore the slough's cultural significance and with proper drainage, the livelihood of pedestrians, drivers, and businesses can be improved. We acknowledge the heat island effect and plan to add green spaces to the area as there are benefits to one's mental health and well-being.[11] Additionally, the improved water quality will help improve ecosystem health like fish habitats.

Global, Cultural, Social

Ideally, residents utilizing the Columbia Slough area should be able to enjoy the river without having to worry about toxic pollutants and unsafe bacteria. Reducing the pollution entering the slough could help rectify this concern and increase the quality of life in the area. Preserving the slough will further deepen the cultural connections that people may have to the slough, especially indigenous nations. We would also like to involve the local community in the planning process to ensure equitable design. Since these facilities are being constructed in their neighborhoods, it is essential that residents and businesses are on board with the interventions, understand their purpose and functioning, and understand the potential maintenance.

Environmental

Implementing stormwater treatment facilities throughout the basin area will help to prevent pollutants from reaching the slough, consequently mitigating damage to the local ecosystem. The establishment of GSI alleviates the heat island effect, sequesters carbon dioxide, and improves air quality. [12] Overall, implementing GSI has several environmental benefits and would be a great addition considering that there is little to no green space in the basin area.

Economic

Our main economic design consideration is to meet regulatory requirements while keeping costs to a minimum. Sourcing manufacturing technologies within the Portland or Pacific Northwest region is ideal for minimizing costs and procurement time. We expect our treatment facilities to be a long-term solution and is easy to maintain. A design will be chosen based on its economic feasibility.

Project Constraints

The design of the stormwater facilities must be approved by the City of Portland. Construction of the stormwater facilities must be within the boundaries of the right-of-way (ROW). The design must ensure that the construction of the project does not impede the local community. This is especially important in terms of construction equity because long-term projects can disturb the existing conditions and create disruptions in the surrounding communities. The design also needs to be within a budget comparable to similar projects established by the City of Portland, which are typically under $2 million.

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[11] Barton, Jo, and Mike Rogerson. "The importance of greenspace for mental health." *BJPsych international* vol. 14,4 79-81. 1 Nov. 2017, doi:10.1192/s2056474000002051

[12] Feinstein, Henry. Rep. Getting to Green: Connected Spaces for Environmental Justice and Stormwater Management at Sayre High School.

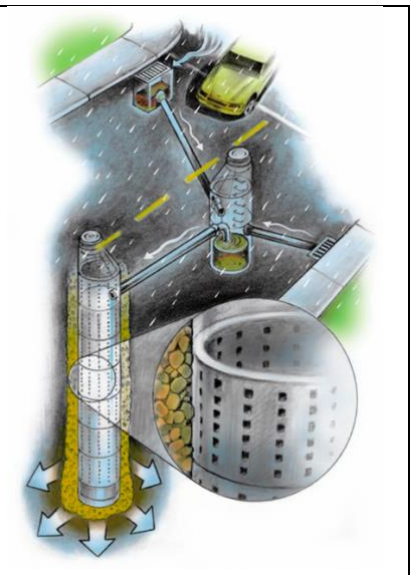
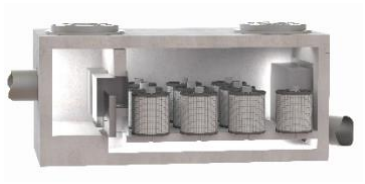
Design Approach

Conceptual Design

To start the design of the stormwater treatment facility for Outfall 65, the team conducted background research on potential stormwater treatment facilities and collected GIS data. All applicable stormwater treatment facilities for the ROW, as per the City of Portland Stormwater Management Manual, were tabulated in a chart (Appendix D). A simplified table highlighting notable facilities is shown in Table 1.

Table 1. Consolidated stormwater facility comparison with descriptions and graphics. A detailed version is provided in Appendix D.

Facility	Description	Graphic
Permeable Pavement	Pervious surfaces that allow infiltration. Usually used in sidewalks, parking lots and low-trafficked roadways. Requires continuous maintenance; Price ranges from $7-13 per sq. ft. of asphalt, and $8-16 per sq. ft. of concrete.	Source: City of Portland
Rain Gardens	Shallow depression of plants that collect, store, and allow precipitation to absorb into the ground or route to a flood network.	Source: City of Portland
Green Street	Combination of several GSI like curb extensions, street planters, rain gardens, etc. Environmental and community benefits.	Source: City of Portland

Sump	Basin structure designed to collect runoff and allow the water to infiltrate into the surrounding soil	 Source: Portland Stormwater Management Manual 2020
Manufactured Stormwater Treatment Technologies (MSTT)	The City of Portland has designated a list of approved proprietary systems in the Stormwater Management Manual. This encompasses technologies such as structured detention and vaults (see photo to the right), among others.	 Source: OldCastle Infrastructure

Design considerations such as cost, size, maintenance, environmental and equity impacts, requirements, and specifications were researched for each facility. Based on the associated information for each facility, the team was able to select approaches that would be the most suitable for the project's goals using a design decision matrix. For instance, a permeable pavement roadway application was ruled out since it would not suit the highly trafficked area of interest and the maintenance needs and costs outweighed the treatment benefits. Additionally, we chose not to include sumps in our design since they primarily address sediment removal and have limitations in terms of treatment. ArcMap and Portland Maps were utilized in the conceptual brainstorming process to map out the locations of each facility. Rough conceptual design ideas were pitched by team members, finalized, and sent to our industry and faculty advisors for feedback.

Stormwater Management Model

A hydrologic and hydraulic model was developed with SWMM v5.2 to predict design flows. Finding hydrologic and hydraulic information about the site is necessary to determine the capacity of the site and sizing of proposed treatment facilities. The model's inputs were gathered from the attribute tables from geospatial layers provided by the City of Portland and the Portland Stormwater Management Model. For the first development of the model, key parameters including the sub-catchment areas, impervious ratios, pipe dimensions, and pipe elevations were provided by BES and analyzed using ArcMap. Unknown parameters for which BES did not provide were set to the default settings. The

stormwater management manual also lists design storm depths, which were simulated in the model (Table 2). To meet water quality standards, 90% of the average annual rainfall must be treated from the site's impervious areas, which equates to what is called the "Water Quality Design Storm" of 1.61 inches.

Table 2. Portland Stormwater Management Model (2020) design storm depths for a 24-hour duration.

Storm Event	24-hour Rainfall Depth [in]
Water Quality	1.61
2-year	2.4
5-year	2.9
10-year	3.4
25-year	3.8

The model consists of 26 sub-catchments, 50 junction nodes, and 51 conduits (Figure 2). The total catchment area of the Outfall 65 basin is 16.4 acres, 45% of which is impervious. The average subbasin is approximately 28,500 square feet per basin, with impervious percentages ranging from 0% to 90%. There are several pipes transporting stormwater, with the smallest pipe being 8 inches in diameter and the largest pipe being 9 feet in diameter, which leads to the two outfalls. The basin drains into two separate outfalls which model a double barrel pipe scenario.

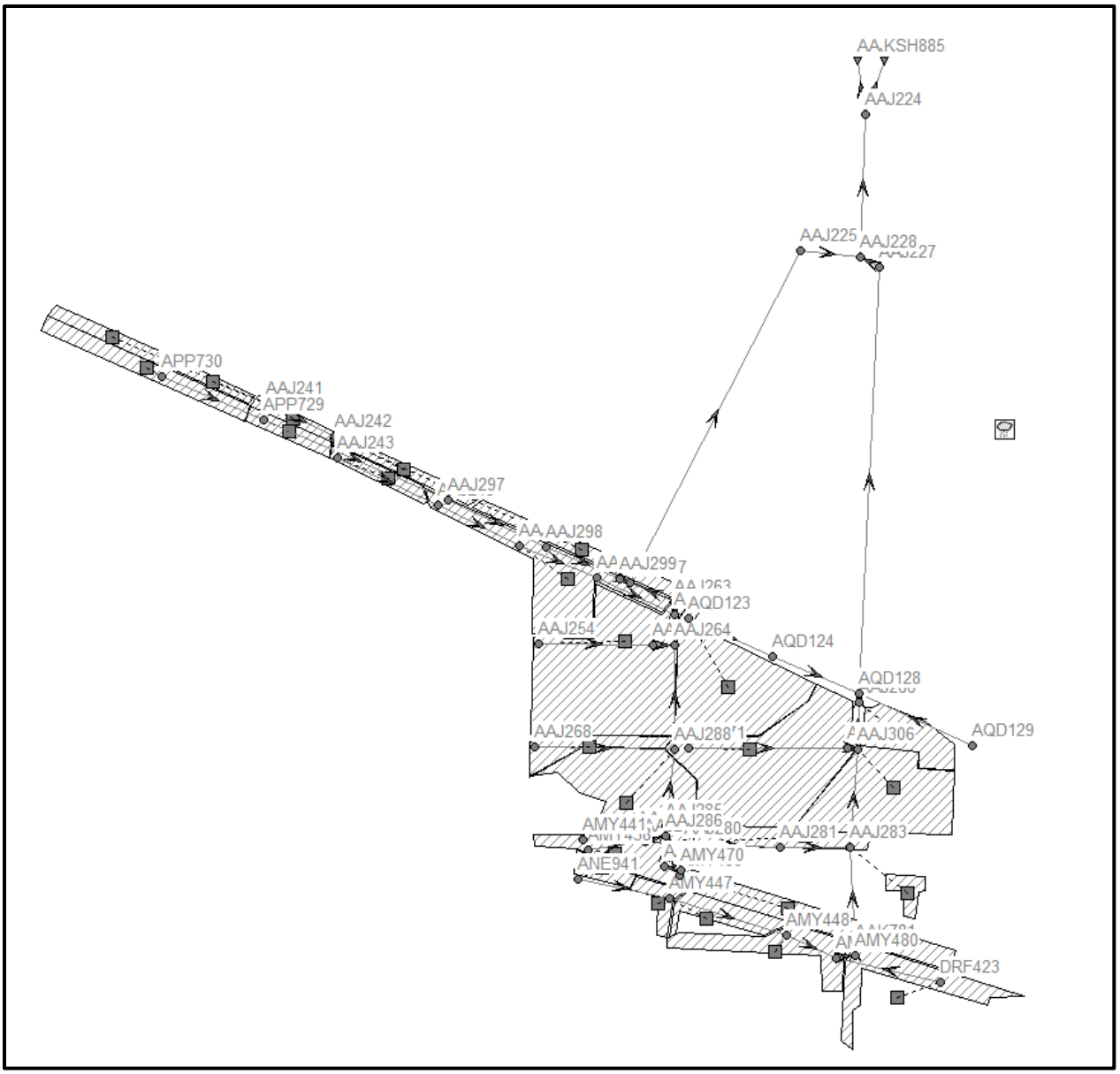


Figure 2. Outfall 65 Basin SWMM map.

Development & Analysis

The Outfall 65 basin model was simulated for 2-year, 5-year, 10-year, 25-year, and Water Quality design storm over an interval of 24-hours. SWMM outputs summary tables of conduit profile graphs, flows through each conduit, surcharging and flooding data, and an outfall loading summary. The outfall loading for the design storms are shown in Table 3.

Table 3. Outfall loading summary of the simulated design storms. The average and maximum flows were obtained by adding the two outfalls together.

Recurrence Interval	Combined Average Flow [cfs]	Combined Maximum Flow [cfs]	Total Volume [million gal]
2-yr	0.62	1.00	0.329
5-yr	0.70	1.19	0.515
10-yr	0.8	1.37	0.579
25-yr	0.86	1.51	0.629
Water Quality	0.46	0.70	0.329

The Portland Stormwater Manual recommends the 10-year, 24-hour storm as the design storm for managing runoff volume. The model was simulated, quantifying the flow going through two conduits of interest, M49932 and M47480, which capture 51% and 37% of the basin area respectively (Figure 3).

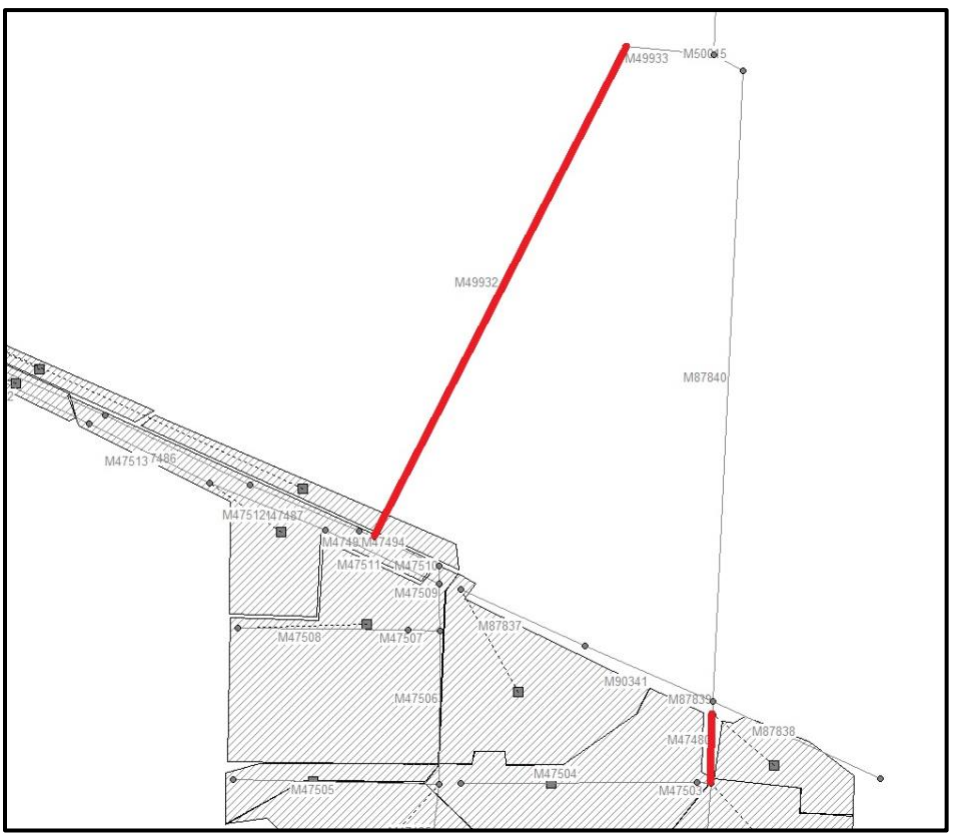


Figure 3. Conduits M49932 (left) and M47480 (right). Flows through these conduits were quantified.

Table 4. Conduit summary of a 10-year, 24-hour design storm and water quality design storm.

Conduit	Maximum Flow [cfs]
10-year Design Storm	
M49932	1.04
M47480	0.70
Water Quality Design Storm	
M49932	0.56
M47480	0.46

Further Analysis

The developed model must be validated for accuracy. Flows through the basin for the 10-year design and water quality design storm were compared with the flows given by BES (Table 5).

Table 5. Comparison of results between our model and BES.

Conduit	Maximum Flow [cfs]	Maximum Flow, BES [cfs]
10-year Design Storm		
M49932	1.04	5.4
M47480	0.70	5.1
Water Quality Design Storm		
M49932	0.56	1.1
M47480	0.46	0.8

Further analysis of the difference in flows is necessary because the simulated flows are smaller than the expected flows. The model can be fine-tuned by adjusting the default parameters or correcting potential routing errors.

Alternatives Analysis

Five design approaches are proposed for Outfall 65 stormwater treatment. The first three designs are each based on one of three main design considerations: economic, environmental consciousness, and equity. Each of the three designs maximized their respective consideration. Combinations 1 and 2 encompass aspects of each of the three primary designs.

Evaluating Criteria

The primary criteria for this design project constitute a successful project and are outlined below:

- ***Water Quality performance.*** Our design must meet stormwater treatment standards of the Portland Stormwater Management Manual (2020 ed.), which is to "achieve 70% TSS removal from the runoff resulting from 90% of the average annual rainfall."
- ***Cost.*** We will take into consideration multiple stormwater treatment technologies and their associated costs. We are striving to design and construct the cheapest option that meets design requirements.

The secondary criteria for this design project are highly desirable, but not essential to the project's overall success:

- ***Environmental Equity.*** Our design strives to incorporate vegetation and green spaces, thus reducing the urban heat island effect onsite, among other benefits such as air quality and habitat creation.
- ***Constructability***. Construction of our facilities will follow the best management practices outlined by the EPA (EPA, 2023). Factors to consider include project footprint, traffic control, construction access, property impacts, and existing site conditions.
- ***Social Equity***. Our proposal aims to minimize any negative impacts that new construction and grey infrastructure can have on local community members' quality of life. Features such as sidewalks and ADA ramps, as well as opportunities for community engagement, contribute to this criteria. Additionally, we intend to address inequities related to land-surface heat vulnerability caused by the urban heat island effect with our design.

Economic Design

The first alternative of the Outfall 65 Water Treatment Project is the most cost-effective approach. For this design, two vaults like the one found in Appendix F will be placed on NE 13th Ave and NE 11th Ave (Figure 4). These vaults will treat the water quality storm event and convey the flow of a 25-year storm event. A vault is a form of gray infrastructure placed underground to treat stormwater. This design uses strictly gray infrastructure to treat the stormwater runoff.

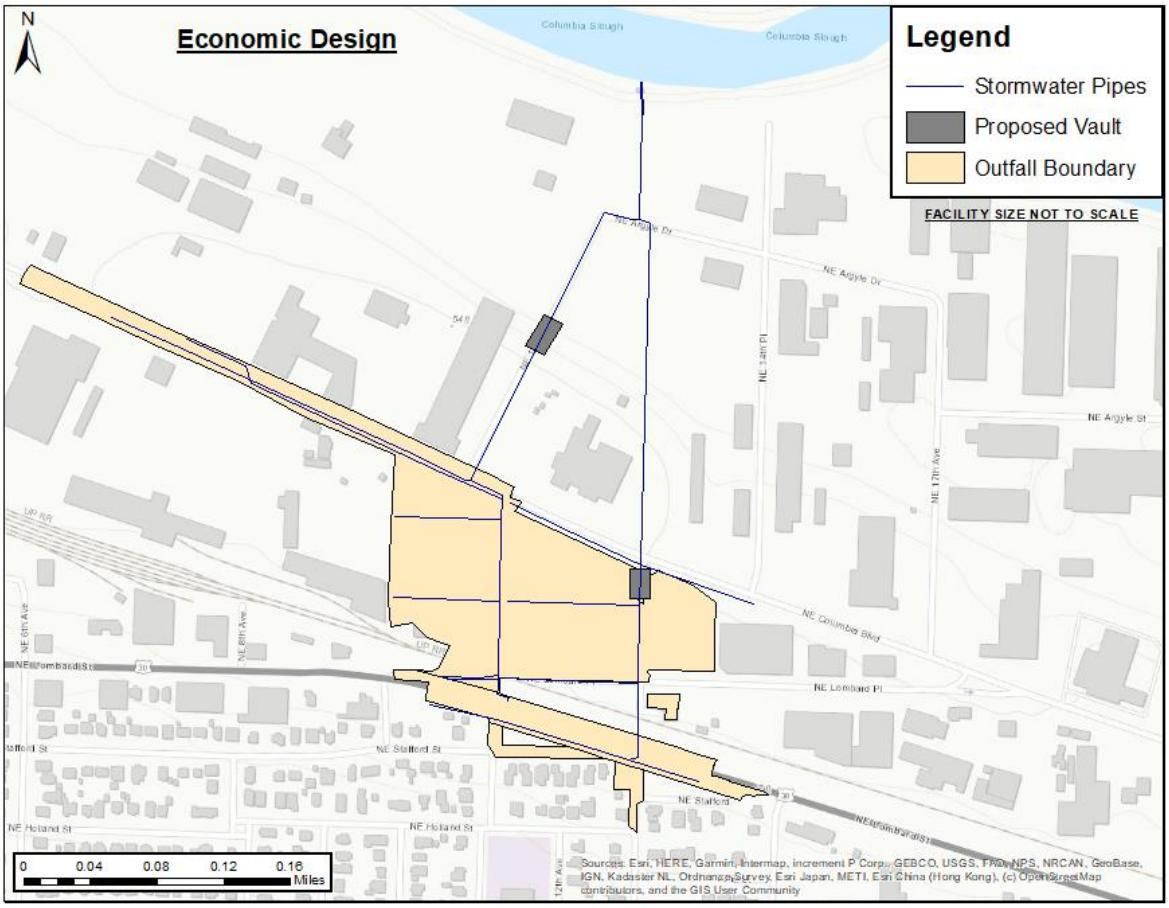


Figure 4. Economic design, focus on cost-effectiveness

When installing vaults, it is required that a sedimentation manhole is placed upstream of the vault to remove any debris and oil that would enter the vault (Appendix F). Placing a sedimentation manhole upstream of the NE 11th Ave vault will be required as well. Due to high traffic on Columbia Blvd and high number of existing utilities in the surrounding areas, placing any treatment facility on this road will prove to be inconvenient for construction and the community, despite being able to treat more stormwater runoff.

The primary advantages of this design include: (1) Performance: These vaults will be able to treat 88% of the Outfall 65 basin area. Placing a vault on the northern end of NE 11th Ave covers 51% of the site and the vault on NE 13th Ave captures 37% of the site. (2) Constructability: NE 11th Ave is currently not paved which will allow for an easier installation of the vault.

This design results in a full gray infrastructure and economically focused alternative, with the following drawbacks: (1) Will not treat 12% of the watershed area; (2) provides no green infrastructure; (3) provides no social equity.

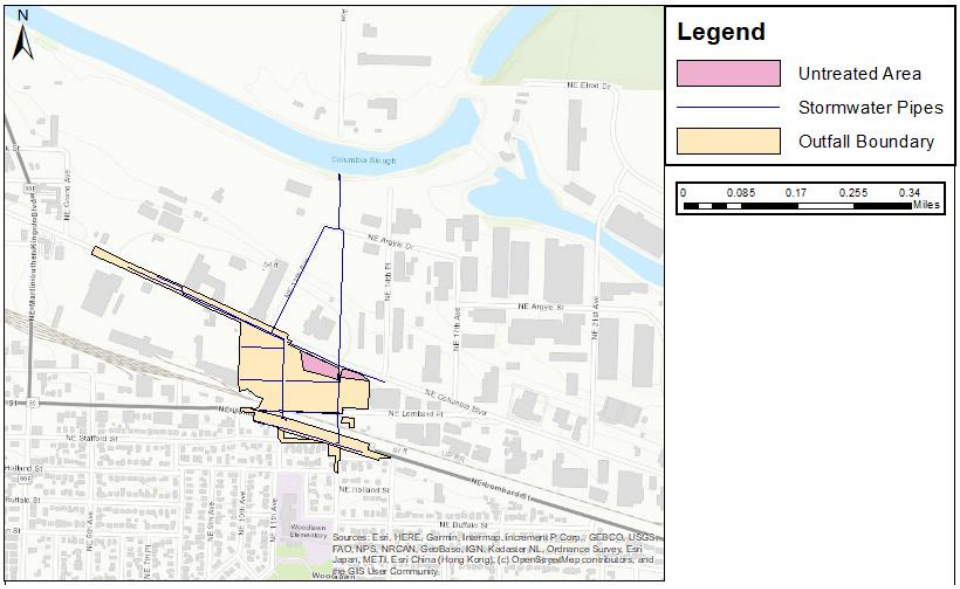


Figure #. Map of untreated basin area.

Green Design

The second alternative of the Outfall 65 Water Treatment Project is centered around GSI. The purpose of this design was to maximize the use of green infrastructure to treat stormwater without the use of grey infrastructure. The design consists of the following: a green street on NE 11th Ave, tree planters on NE Russet St, and bioretention cells and planter bump outs scattered throughout (Figure 5). The green street will feature a permeable sidewalk made of paver stones, along with planters.

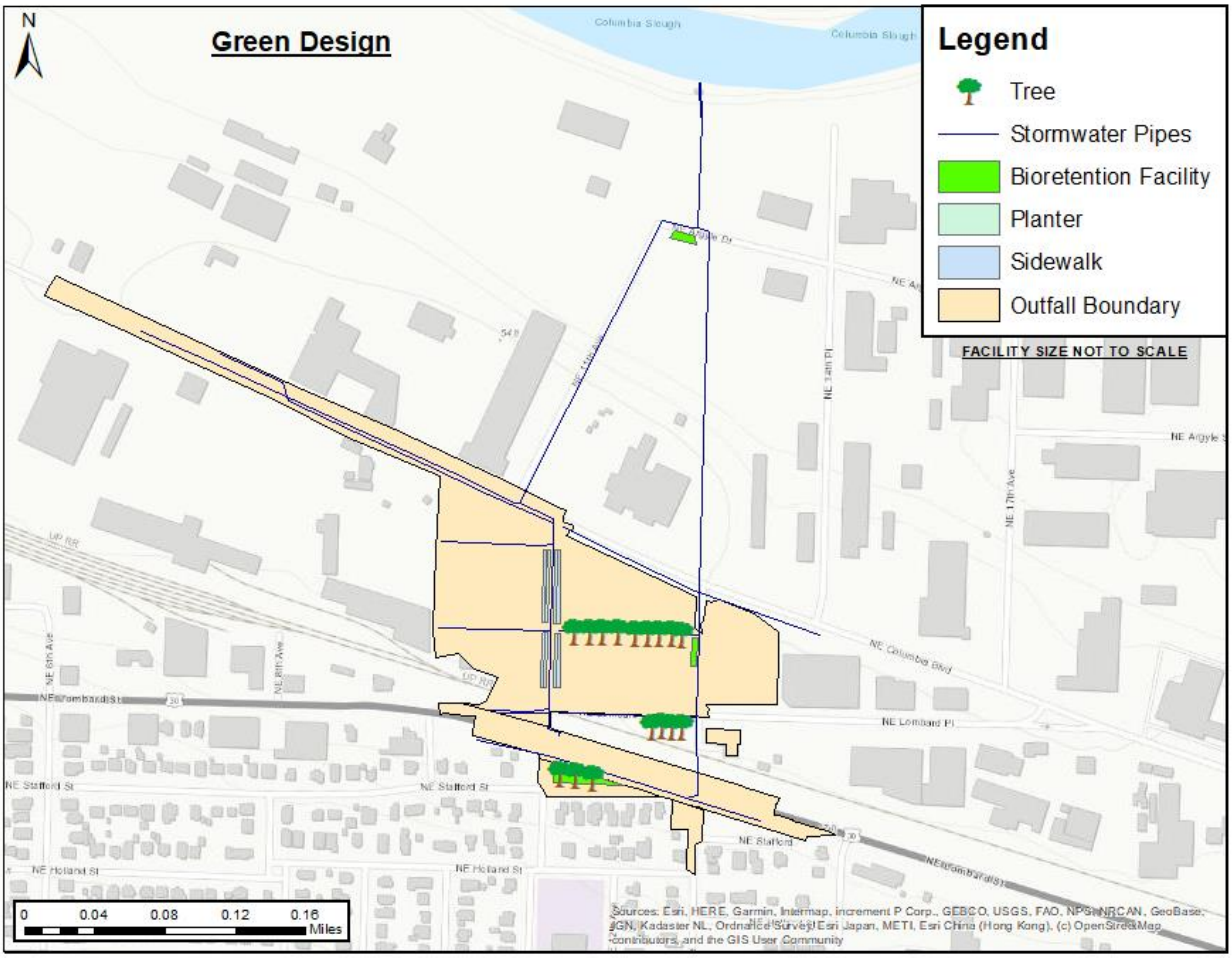


Figure 5. Green Design, focus on green infrastructure

Green streets are vegetated stormwater facilities located on sidewalks that capture street runoff (Figure 6). Tree planters can also be incorporated into sidewalks, contributing to stormwater collection while adding vegetation to urbanized areas. Bioretention cells or rain gardens give similar benefits to green streets but can be used in larger areas. The soil media in bioretention cells can be amended to enhance performance. Biochar, which is the byproduct of the pyrolysis of biomass, has been found to have high water retention and metal removal properties making it an attractive amendment to the soil media in bioretention cells.[13] Metal pollution of runoff from this basin is of special concern, due to the highly industrial nature of the site. Therefore, this design proposes biochar-amended bioretention cells. See Appendix E for visualizations of the mechanisms behind different GSI interventions.

[13] Minnesota Pollution Control Agency. *Minnesota Stormwater Management Manual*. Minnesota Pollution Control Agency, 2023.

Figure 6. Green Street (Source: Friends of Green Streets)

While the green design is a more natural alternative, it mainly focuses on surface stormwater runoff and reducing the amount of water flowing into the basin. This means that water flowing into the basin not from the locations of the green infrastructure will not be as affected. Due to the existing utilities and highly trafficked nature of Columbia Blvd. and Lombard St., the areas in which GSI can be implemented are limited. Therefore, GSI alone would not be able meet the performance standards for stormwater treatment.

The primary advantages of this design include: (1) Aesthetics: the added vegetation and green spaces will help beautify the area, (2) Urban heat reduction: evapotranspiration from vegetation in the GSI will increase cooling, (3) Pedestrian mobility: the sidewalk in the green street will make it easier for pedestrians to access the area.

The primary drawbacks include: (1) Performance: this design would only treat no more than 15% of the basin area (area may be smaller since the GSI would only treat some surface areas), therefore failing to meet the objective of this project, (2) Cost: GSI is more expensive than grey infrastructure.

Equitable Design

The Equitable Design is centered around social equity. The purpose of this design was to provide the maximum benefit to the community.

This design features some aspects of the Green Design, since green spaces provide many social benefits in addition to environmental, as well as the Economic Design due to the social benefits associated with runoff reduction. For this reason, this design preserved the green street on NE 11th Ave., as well as the trees on NE Russet St., although this design alternative calls for less, more sparsely spread trees to accommodate the existing encampments of unhoused people. The vault on NE Russet and NE 13th Ave, will also be incorporated in this design (Figure 7).

Features exclusive to the equitable design include a two-part GSI education area ranging from Woodlawn Elementary School to an existing tax lot at the intersection of NE Lombard and NE Stafford St. This includes permeable pavement walkway with educational panels in the place of the existing sidewalk on NE 11th Ave ranging from the school to NE Stafford St. The walkway leads directly to an educational

bioretention facility “park” on the tax lot. The “park” contains biochar-amended rain gardens, along with a permeable pavement walkway lined with more educational panels and a few benches.

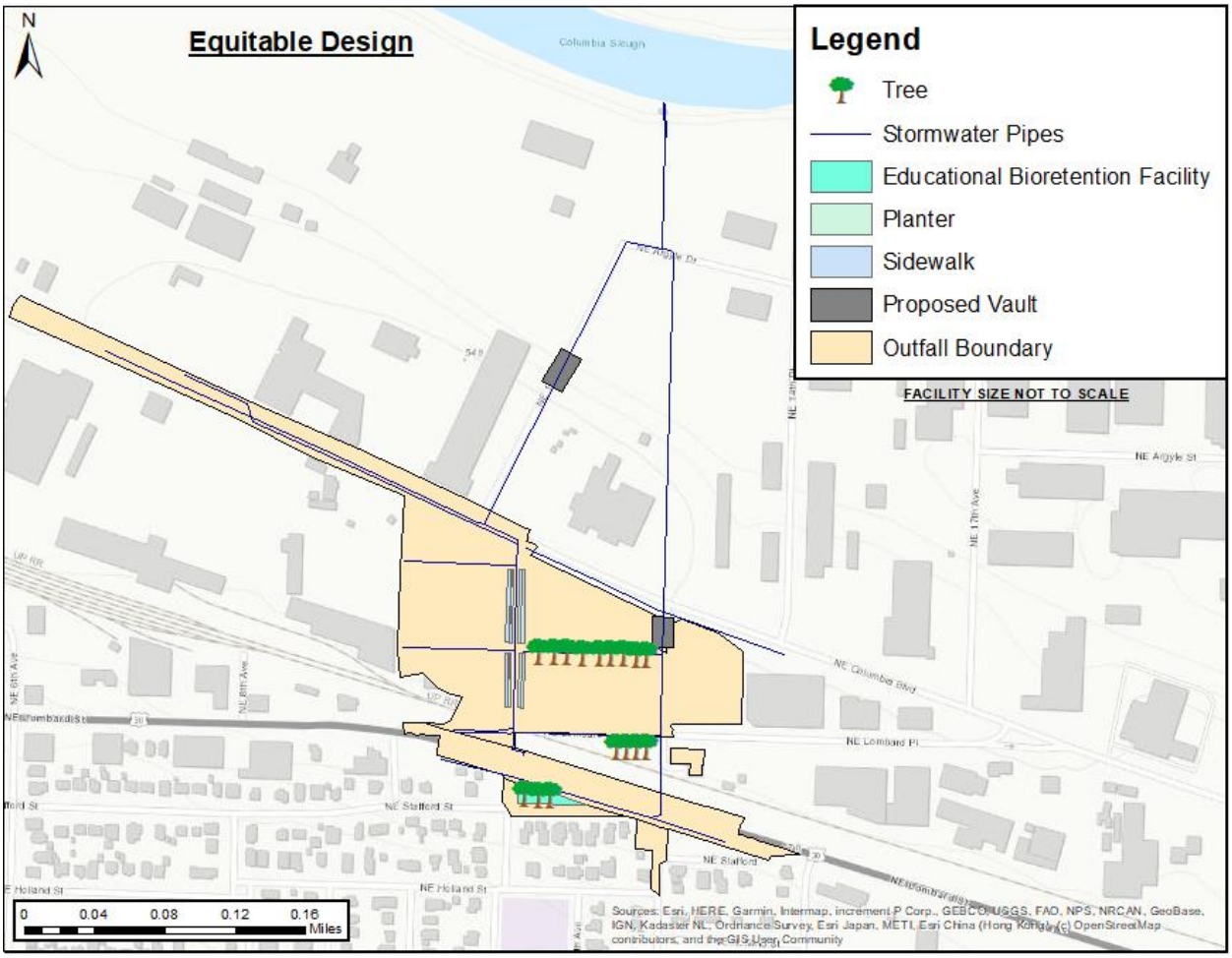


Figure 7. Equitable Design, focus on social equity

This educational area would serve the elementary school and promote community engagement by providing students with an opportunity to help maintain the facilities. For this reason, the GSI facilities incorporated in this design will be designed as per the specs in the Portland Stormwater Management Manual. Pre-manufactured solutions such as Contech’s Filterra rain-gardens require properly trained maintenance personnel for upkeep, which would negate the opportunities for community involvement.

The primary advantages of this design include: (1) Performance: this design would be able to meet the performance standards for treatment for the site, (2) Education: the educational bioretention facility would provide opportunities to learn about stormwater management and help maintain the facilities, (3) Community engagement: The stormwater park would provide a place of leisure for the surrounding community, (4) Aesthetics: the added vegetation and green spaces will help beautify the area, (5) Urban heat reduction: evapotranspiration from vegetation in the GSI will increasing cooling, (6) Pedestrian mobility: the sidewalk in the green street will make it easier for pedestrians to access the area.

Drawbacks associated with this design include: (1) Cost: GSI is more expensive than grey infrastructure, (2) Constructability: proximity to highly residential zones, existing infrastructure such as sidewalks.

Combination 1

Combination 1 represents a maximalist approach. It consists of two vault systems placed on NE 13th Ave. and NE 11th Ave., a green street on NE 11th St., tree planters on NE Russet St., and the educational tax lot (Figure 8gure 8).

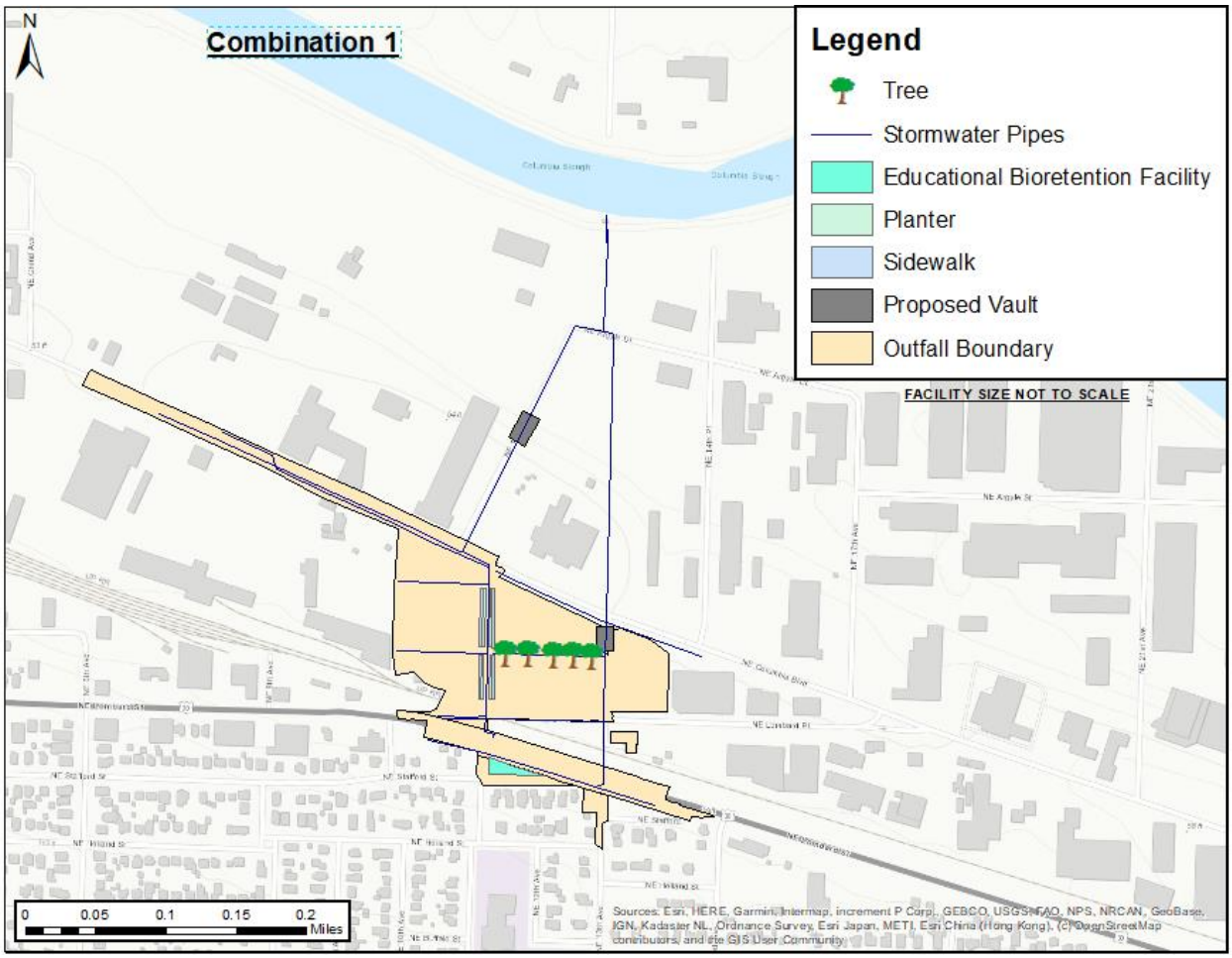


Figure 8. Combination 1

The primary advantages of this design include: (1) Performance: this design would be able to meet the performance standards for treatment for the site, (2) Education: the educational facility would provide opportunities to learn about stormwater management and help maintain the facilities, (3) Community engagement: The stormwater park would provide a place of leisure for the surrounding community, (4) Aesthetics: the added vegetation and green spaces will help beautify the area, (5) Urban heat reduction: evapotranspiration from vegetation in the GSI will increasing cooling, (6) Pedestrian mobility: the sidewalk in the green street will make it easier for pedestrians to access the area.

The primary drawback of this design is the high associated cost.

Combination 2

Combination 2 is a simplified version of Combination 1. It encompasses what the Team decided were the most critical components of the economic, equitable, and green designs. This design consists of two vaults placed on NE 11th and NE 13th Ave. and a green street on NE 11th Ave (Figure 9e 9).

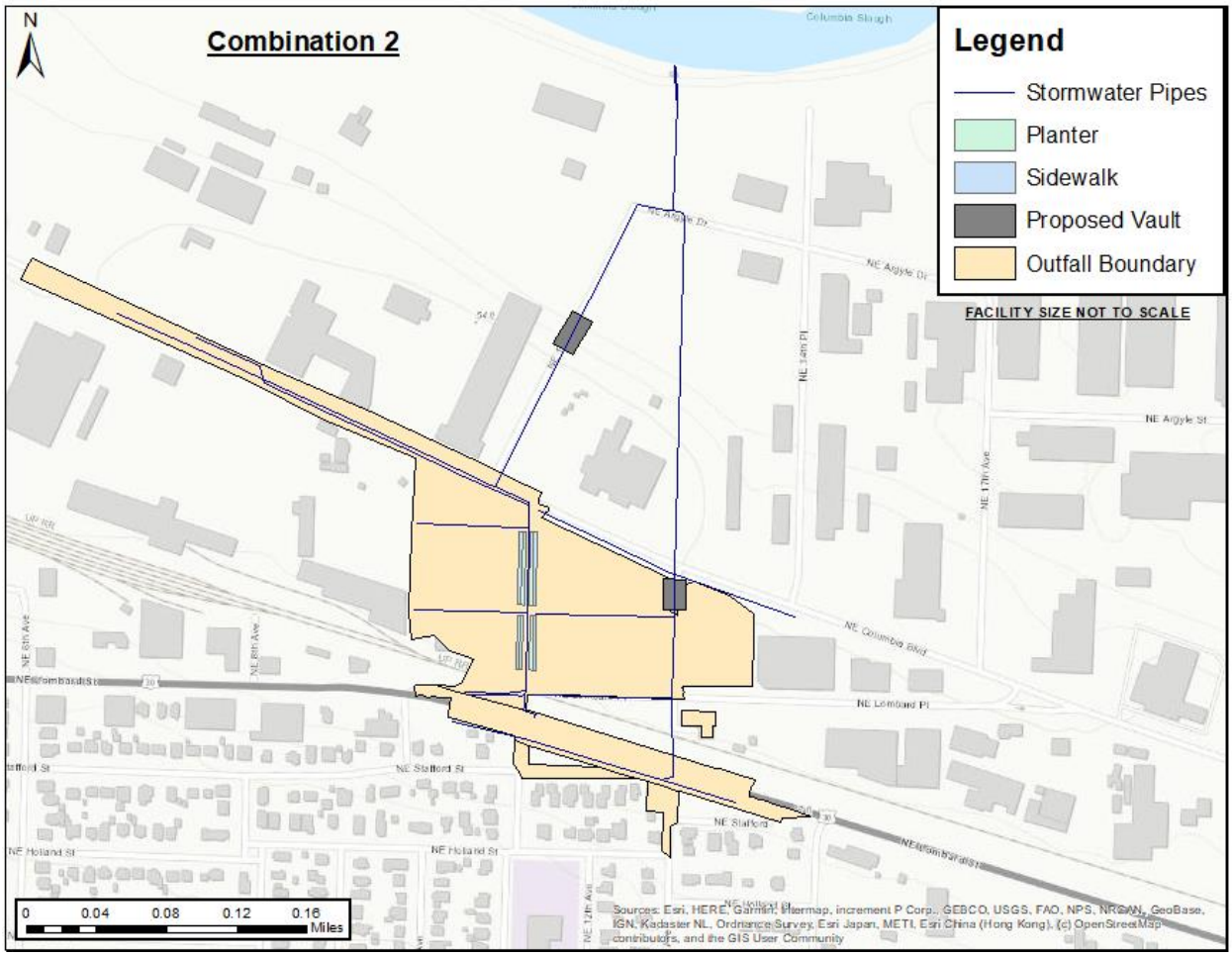


Figure 9. Combination 2

The primary advantages of this design include: (1) Performance: this design would be able to meet the performance standards for treatment for the site, (2) Urban heat reduction: evapotranspiration from vegetation in the GSI will increasing cooling, (3) Aesthetics: the added vegetation and green spaces will help beautify the area, (4) Pedestrian mobility: the sidewalk in the green street will make it easier for pedestrians to access the area.

The primary drawback of this design is cost, although it would be significantly more cost-effective than Combination 1.

Design Evaluation and Recommendation

A design matrix was developed based on the design considerations. In this matrix, each design was given a score in terms of how well it meets the given consideration (a score of 0 represents no contribution, 4 indicates maximum contribution). The primary considerations (water quality performance and cost) were given more weight than the secondary considerations (environmental equity, social equity, and constructability), reflecting their importance in the design. The five designs were evaluated on five different criteria: performance, cost, environmental equity, social equity, and constructability.

- **Water Quality Performance (Weight = 1)**: ability to meet stormwater treatment quality standards.
    - 0: does not meet water quality standards
    - 4: meets the standards
- **Cost (Weight=1)**: cost of materials, construction, and other associated fees
    - 0: highly expensive
    - 4: highly cost-effective
- **Environmental Equity (Weight=0.5)**: amount of vegetation and green space in the design, includes green streets, trees, planters, etc.
    - 0: no added vegetation
    - 4: maximum implementation of greenspace
- **Social Equity (Weight = 0.5)**: benefit to the surrounding community, for example: pedestrian safety, urban heat reduction, educational opportunities, etc.
    - 0: no social benefit
    - 4: maximum social benefit
- **Constructability (Weight = 0.25)**: ease of construction determined by the proximity of facilities, ease of access, existing site conditions, etc.
    - 0: highly difficult to construct
    - 4: simple construction

Table 66. Design Decision Matrix. A score of 0 represents no contribution to the consideration, and 4 represents the maximum contribution.

Considerations	Wt.	Economic	Green	Equitable	Combo 1	Combo 2
Water Quality Performance	1	4	0	4	4	4
Cost	1	3	1	0	0	2
Environmental Equity	0.5	0	4	3	4	3
Social Equity	0.5	0	3	4	4	3
Constructability	0.25	3	2	0	0	3
Total		7.75	5	7.5	8	**9.75**

First Place: Combination 2

Combination 2 features maximum performance with high scores for environmental consciousness and constructability. It provides a blend of grey and green infrastructure reaping benefits from both

approaches. The higher costs associated with GSI are mitigated by limiting the number of such facilities throughout the design yet preserving enough to maintain the environmental and social benefits. Therefore, this would be the lowest-cost design that still incorporates GSI and provides the associated advantages.

Score breakdown:

- Water Quality Performance: 4, meets requirements
- Cost: 2, limited GSI
- Environmental Equity: 3, green street
- Social Equity: 3, sidewalks, ADA ramps, and cooling (from vegetation)
- Constructability: 3, relative concentration of facilities, limited existing infrastructure
- **Total:** 9.75

Second Place: Combination 1

In Combination 1, the added educational area and green infrastructure (GSI) in the design yields maximum contributions to environmental and social equity. Coupled with grey infrastructure in the form of vaults, the blend of facilities in this design would generate high treatment performance, as well as runoff reduction. However, due to the high amount of GSI, this design would be one of the most expensive alternatives, outside of the Equitable design. Additionally, the diversity of the facilities and their geographic distance (as opposed to facilities in close vicinity) would make construction more challenging.

Score breakdown:

- Water Quality Performance: 4, meets requirements
- Cost: 0, lots of GSI, educational stormwater park
- Environmental Equity: 4, green street, trees, bioretention
- Social Equity: 4, sidewalk, ADA ramps, cooling (from vegetation), educational stormwater park
- Constructability: 0, facilities are scattered, existing infrastructure
- **Total:** 8

Third Place: Economic Design

The Economic design features two vaults only, and scores well for the primary considerations of cost and performance, as well as constructability reflecting the practical nature of the design. However, it contributes nothing to environmental and social equity.

Score breakdown:

- Water Quality Performance: 4, meets requirements
- Cost: 3, no GSI
- Environmental Equity: 0, no added greenspace
- Social Equity: 0, no community benefit
- Constructability: 3, only vaults
- **Total:** 7.75

Fourth Place: Equitable Design

The strengths of the Equitable Design are environmental consciousness and social equity. It faces low scores for constructability, due to the work to be done in residential areas and the distance between different facilities, as well as performance since this configuration would not be able to treat all the runoff in the basin. The educational areas combined with the high amount of GSI would make this one of the more expensive options, as well.

Score breakdown:

- Water Quality Performance: 4, meets requirements
- Cost: 0, lots of GSI, educational stormwater park
- Environmental Equity: 4, green street, trees, bioretention
- Social Equity: 4, sidewalk, ADA ramps, cooling, educational areas, park
- Constructability: 0, facilities are scattered, existing infrastructure
- **Total:** 7.5

Last Place: Green Design

Despite high ratings in terms of environmental consciousness and social equity, GSI alone would not be able to treat all the runoff to standard. Additionally, this approach would be more expensive and more challenging in terms of constructability too due to their geographic distribution, and existing site conditions.

Score breakdown:

- Water Quality Performance: 0, does not meet requirements
- Cost: 0, lots of GSI
- Environmental Equity: 4, green street, trees, bioretention
- Social Equity: 4, sidewalk, ADA ramps, cooling
- Constructability: 0, facilities are scattered, existing infrastructure
- **Total:** 5

Future Work

Over the course of the Fall 2023 semester, our team has completed the first half of our project, consisting of background research, model development, and conceptual design. In the Spring, we will commence the final half of the project, which will encompass design to 60% completion, including drafts, submittals, and other necessary construction documents.

For our next steps, we will first refer to our alternatives analysis and settle on a final proposed design. The SWMM model will also be further refined by adjusting default parameters and fixing potential routing errors. Then, we will commence our design process based on the facilities proposed in said alternative. In a second site visit, we will assess existing site conditions through infiltration testing and take survey measurements. This will help to reassess our chosen design and amend it as needed, as well as allow us to begin drafting our plans in AutoCAD.

Another significant aspect of the design process is reaching out to manufacturers about the specific products we intend to incorporate. By sending them flow data calculated from our model, they will be able to provide us with correctly sized facility specs that we can integrate into our plans, as well as associated cost estimates.

Appendices

Appendix A: Teamwork and Professionalism

Table A7. Leadership and Roles

Team Member	Roles and Description
Gus Shelley	Design Lead – leading conceptual design process and the vision of the project Technical Drafter – responsible for modeling and drafting (ArcMap, AutoCAD) Budget – Planning and creating budget for the project
Bella Small	Researcher – conduct background research, crafting alternatives analysis, and design matrix Note-taker – records meeting minutes and discussion notes Budget – planning and creating budget for the project
Maya Struzak	Design Lead – leading conceptual design process and the vision of the project Outreach Coordinator – providing timely responses to emails to faculty and industry advisors and organizing meetings Researcher – conduct background research, crafting alternatives analysis, and design matrix Budget – planning and creating budget for the project
Jayson Tagala	Group Facilitator – ensuring that group synergy is appropriate, delegating tasks when necessary, and reminding group of deadlines Technical Drafter – responsible for modeling and drafting (ArcMap, SWMM, AutoCAD) Scribe – responsible for setting up documents, managing class deliverables, editing and proofreading

Team logistics were executed fairly and in a professional manner. The following forms of communication were used:

- Email. Used to contact advisors.
- Microsoft Teams. Organize files, notes, and meetings.
- Group Message. Chat designated for team members to discuss deliverables, tasks, etc.

Meetings were scheduled weekly, with industry advisor meetings at 2:30pm on Mondays, and faculty advisor meetings at 3:00pm on Wednesdays. Industry advisor meetings were held via Microsoft Teams and were used to review materials, ask questions related to the project, and share updates. Faculty meetings were conducted in-person, and allowed team members to check-in and share updates. Team member meetings occurred at 5:00pm on Thursdays. Decisions were made on a majority-vote basis.

There were some aspects of the project that were not covered in the University's engineering curriculum. Further research on treatment facilities, stormwater management and design, watershed modeling experience, ArcMap, and project management were necessary. The following modern engineering tools were used:

- ArcMap
- Portland Maps
- EPA SWMM
- Microsoft Office Applications (Teams, Project, etc.)

Appendix B: Meeting Minutes and Meeting Agendas

Meeting Date	Industry Advisor	Faculty Advisor
Week 1 (8/27 – 9/2)		Project Introduction and Overview Logistics and Planning
Week 2 (9/3 – 9/9)	Kick Off Event [Review] Background Materials, Plans, Expectations	
Week 3 (9/10 – 9/16)	First Meeting Intro to ArcMap, Stormwater Management Manual, Approved Technologies	[Review] Team Charter Document Start Background Research [Q&A] Pollutant loading, site testing, computer modeling
Week 4 (9/17 – 9/23)	[Q&A] Constraints, Construction Processes, Budgets, Flow Calculations	
Week 5 (9/24 – 9/30)	[Q&A] GSI & Equity, Using traditional vs. GSI, Pervious Pavement	[Review] Project Scope Document Preparation for site visit Learning to use EPA SWMM
Site Visit (9/28)	Visit to OF65 basin area, discussed treatment options, map discrepancies, area outside of basin	
Week 7 (10/8 – 10/14)		Begin facility comparison matrix [Q&A] SWMM Progress, ArcMap attributes, Green Streets, Permeable Pavement
Week 8 (10/15 – 10/21)		
Week 9 (10/22 – 10/28)	[Q&A] SWMM, Conceptual Design Process, Reaching out to MSTT manufacturers	[Q&A] Gantt chart, SWMM Progress, Possibility of driveway center strips and planters
Week 10 (10/29 – 11/4)	[Review] Conceptual Design v1	[Review] Conceptual Design v2
Week 11 (11/5 – 11/11)	[Review] Conceptual Design v3, suggestions to green design [Q&A] No Access Points on ArcMap	[Review]
Week 12 (11/12 – 11/18)	[Review] Conceptual Design v4 [Q&A] Estimated flows	[Review] Final Deliverable to Faculty
Week 13 (11/19 – 11/25)		
Week 14 (11/26 – 12/2)	[Review] Final Poster, Map Updates, Plans for next semester, Design Matrix adjustments	[Review] Final Poster
Week 15 (12/3 – 12/9)		[Review] Final Deliverable

Appendix C: Project Schedule

Task	Duration	Start	End	Predecessors
Semester 1: Fall 2023				
Task 1: Project Management	11 days	Fri 9/8/23	Fri 9/22/23	n/a
Task 2.1: Gather reference handbooks and historical data	26 days	Fri 9/8/23	Fri 10/6/23	Task 1
Deliverable 1: Project Scope & Requirements to UP Advisors	7 days	Fri 9/8/23	Fri 9/15/23	n/a
Deliverable 1.1: Project Scope & Requirements to Industry Advisor	11 days	Fri 9/15/23	Fri 9/22/23	Deliverable 1
Deliverable 1.2: Project Scope & Requirements Final	7 days	Fri 9/22/23	Fri 9/29/23	Deliverable 1.1
Task 2.2: Conduct a Site Visit	1 day	Th 9/28/23	Th 9/28/23	Task 1 Task 2.1
Deliverable 2: Project Report Draft 1 (Outline)	5 days	Mon 10/2/23	Fri 10/6/23	Deliverable 1.2
Task 2.3: Evaluate Treatment Strategies	21 days	Wed 10/4/23	Wed 10/25/23	Task 2.1 Task 2.2
Deliverable 3: Project Plan		Mon 10/9/23	Fri 10/13/23	Deliverable 2
Task 2.4: Development of stormwater management model (SWMM)		Mon 10/9/23	Fri 10/27/23	Task 2.1
Task 3.1: Conceptual Design Draft	5 days	Mon 10/30/23	Wed 11/8/23	Task 2.2 Task 2.4 Deliverable 2 Deliverable 3
Deliverable 3: Project Report Draft 2		Wed 10/30/23	Fri 11/10/23	Deliverable 2 Task 3.1
Deliverable 4: Final Poster		Mon 10/30/23	Fri 11/22/23	Deliverable 3
Deliverable 5: Alternatives Analysis		Mon 11/13/23	Mon 11/27/23	Deliverable 3
Deliverable 6: Final Project Report		Mon 11/27/23	Fri 12/8/23	Deliverable 3 Deliverable 5
Deliverable 7: Global Factors Memo		Fri 12/1/23	Wed 12/6/23	Deliverable 3 Deliverable 5
Semester 2: Spring 2024				
Task 4: Revisit site	1 day	Wed 1/24/24	Wed 1/24/24	Deliverable 6
Task 5: Drafting	28 days	Wed 1/24/24	Fri 3/1/24	Task 4
Task 6: Organize Project Deliverables	9 days	Mon 3/11/24	Thu 3/21/24	Deliverable 6 Task 5
Deliverable 8: Final Report and Oral Presentation	39 days	Thu 3/11/24	Fri 4/19/24	Task 6

Appendix D: Stormwater Management Model Inputs

Table C1. Sub-catchment inputs. 1

Name	Rain Gage	Outlet	Area	%Imperv	Width	%Slope	CurbLen
S10	WQ	APP730	0.439591701	80.56	62.57717154	0.5	0
S11	WQ	APP729	0.197965351	80.67	37.16970134	0.5	0
S12	WQ	AAJ241	0.157121057	93.54	29.37421993	0.5	0
S13	WQ	AAJ242	0.246931697	85.29	28.38085678	0.5	0
S14	WQ	AAJ243	0.238631473	68.86	40.13431258	0.5	0
S15	WQ	AAJ242	0.428034522	82.46	51.36414266	0.5	0
S16	WQ	AMY438	0.492699912	99.59	75.04198652	0.5	0
S17	WQ	AAJ280	0.52646033	27.04	59.41091182	0.5	0
S18	WQ	AAJ306	1.412364247	12.13	214.3644133	0.5	0
S19	WQ	AAJ251	1.03876353	67.29	211.4417727	0.5	0
S2	WQ	AAJ283	0.132410696	0.00	29.42760171	0.5	0
S20	WQ	AAJ254	2.375589562	67.29	236.7978977	0.5	0
S21	WQ	AAJ266	0.523171514	35.12	89.02090297	0.5	0
S22	WQ	AAJ288	1.357017434	37.38	175.4055769	0.5	0
S23	WQ	AAJ268	0.343840211	21.92	42.67145184	0.5	0
S24	WQ	AAJ271	2.131333342	86.16	221.0497152	0.5	0
S25	WQ	AAJ241	0.2117534	20.74	29.85106181	0.5	0
S26	WQ	AAJ241	0.121179313	83.62	31.05041686	0.5	0
S3	WQ	AQD123	1.42268378	89.77	175.0624448	0.5	0
S4	WQ	DRF423	0.617426915	75.72	60.98665852	0.5	0
S5	WQ	AMY448	0.548669135	64.33	84.45239408	0.5	0
S6	WQ	AMY453	0.741047109	69.81	47.68096314	0.5	0
S7	WQ	AMY447	0.353662835	88.68	49.37677275	0.5	0
S8	WQ	ANE941	0.171050858	94.72	53.60413932	0.5	0
S9	WQ	AMY447	0.15020766	90.14	63.52471513	0.5	0

Table C2. Junction inputs. 1

Name	Elevation	MaxDepth	Name	Elevation	MaxDepth
AAJ224	8.81	31.19	AAJ306	36.87	14.13
AAJ225	9.21	12.79	AAK781	46.34	18.89
AAJ227	9.06	13.27	AAK926	43.94	17.06
AAJ228	9.34	12.99	AMY438	46.24	13.91
AAJ241	47.31	6.69	AMY441	46.82	13.21
AAJ242	45.76	8.24	AMY447	56.69	5.02
AAJ243	45.53	8.47	AMY448	55.46	8.71
AAJ245	43.75	7.25	AMY449	54.85	9.87
AAJ251	41.91	7.87	AMY453	56	5.69
AAJ252	39.81	9.89	AMY470	44.29	16.98
AAJ254	47.3	5.7	AMY480	46.34	18.93
AAJ257	35.46	10.31	ANE941	57.64	4.36
AAJ261	44.25	7.25	APP729	43.86	8.48541
AAJ263	35.54	14.96	APP730	44.3	7.28401
AAJ264	36.67	14.71	AQD123	37.5	13.1715
AAJ266	30.23	15.77	AQD124	35	15.2286
AAJ268	53.88	6.12	AQD128	30.22	16.9765
AAJ271	46.14	7.86	AQD129	33	8.3614
AAJ274	43.14	7.86	DRF423	55.83	9.96
AAJ278	46.52	13.38			
AAJ279	45.87	14.03			
AAJ280	45.9	13.1			
AAJ281	44.4	13.6			
AAJ283	41	13			
AAJ285	43.17	16.68			
AAJ286	43.26	16.59			
AAJ288	40.01	13.99			
AAJ292	35.78	14.72			
AAJ297	43.35	7.65			
AAJ298	41.9	7.89			
AAJ299	35.47	14.53			

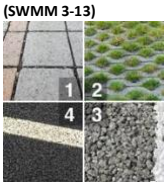
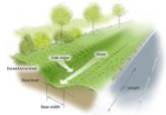
Table C3. Conduit inputs. 1

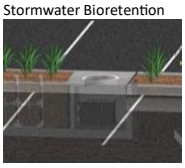
Name	From Node	To Node	Length	Roughness	InOffset	OutOffset	InitFlow	MaxFlow
M47479	AAJ283	AAJ306	246	0.26	0	0	0	0
M47480	AAJ306	AAJ266	114	1	0	0	0	0
M47483	AAJ279	AAJ286	36	0.26	0	0	0	0
M47484	AAJ286	AAJ285	12	1	0	0	0	0
M47485	AAJ285	AAJ288	235	1	0	0	0	0
M47486	AAJ297	AAJ298	299	0.26	0	0	0	0
M47487	AAJ298	AAJ299	207	0.26	0	0	0	0
M47493	AAJ299	AAJ257	16	0.26	0	0	0	0
M47494	AAJ263	AAJ257	110	1	0	0	0	0
M47500	AAJ281	AAJ283	249	1	0	0	0	0
M47501	AAJ280	AAJ281	164	1	0	0	0	0
M47502	AAJ278	AAJ285	60	1	0	0	0	0
M47503	AAJ274	AAJ306	12	0.26	0	0	0	0
M47504	AAJ271	AAJ274	404	0.26	0	0	0	0
M47505	AAJ268	AAJ288	337	0.26	0	0	0	0
M47506	AAJ288	AAJ264	257	1	0	0	0	0
M47507	AAJ261	AAJ264	32	0.26	0	0	0	0
M47508	AAJ254	AAJ261	303	0.26	0	0	0	0
M47509	AAJ264	AAJ292	65	1	0	0	0	0
M47510	AAJ292	AAJ263	13	1	0	0	0	0
M47511	AAJ252	AAJ292	212	0.26	0	0	0	0
M47512	AAJ251	AAJ252	209	0.26	0	0	0	0
M47513	AAJ245	AAJ251	230	0.26	0	0	0	0
M49932	AAJ257	AAJ225	946	1	0	0	0	0
M49933	AAJ225	AAJ228	109	1	0	0	0	0
M49934	AAJ228	AAJ224	329	1	0	0	0	0
M49935	AAJ224	AAJ223	109	0.26	0	0	0	0
M50012	AAJ243	AAJ245	222	0.26	0	0	0	0
M50013	AAJ242	AAJ243	57	0.26	0	0	0	0
M50014	AAJ241	AAJ242	207	0.26	0	0	0	0
M50015	AAJ227	AAJ228	38.29	0.26	0	0	0	0
M50022	AAK926	AAJ286	60	1	0	0	0	0
M57630	AAK781	AAJ283	231.74	1	0	0	0	0
M57631	AMY448	AMY449	139.103927	0.26	0	0	0	0
M57632	DRF423	AMY449	225.099309	0.26	0	0	0	0
M57633	AMY449	AMY480	28.967068	1	0	0	0	0
M57634	AMY480	AAK781	2.58	1	0	0	0	0
M57635	AMY447	AMY448	282.279108	0.26	0	0	0	0
M57640	AMY441	AAJ278	118.594401	0.26	0	0	0	0
M57641	AMY438	AAJ279	134.980758	0.26	0	0	0	0
M63251	AMY453	AMY470	9.368959	1	0	0	0	0
M63253	AMY470	AAK926	37.841817	0.26	0	0	0	0
M73039	AAJ224	KSH885	109.738602	0.26	0	0	0	0
M73727	ANE941	AMY447	219.110205	0.26	0	0	0	0
M87140	APP730	APP729	224.406816	0.26	0	0	0	0
M87141	APP729	AAJ297	498.218977	0.26	0	0	0	0
M87837	AQD123	AQD124	211.975082	0.26	0	0	0	0
M87838	AQD129	AQD128	378.944417	0.26	0	0	0	0
M87839	AAJ266	AQD128	13.571468	0.26	0	0	0	0
M87840	AQD128	AAJ227	1114.444176	0.26	0	0	0	0
M90341	AQD124	AQD128	241.579167	0.26	0	0	0	0

Table C4. Cross-sections inputs. 1

Link	Shape	Geom1	Link	Shape	Geom1	Link	Shape	Geom1
M47479	CIRCULAR	4.5	M47508	CIRCULAR	0.666667	M57632	CIRCULAR	1
M47480	CIRCULAR	4.5	M47509	CIRCULAR	7	M57633	CIRCULAR	1.25
M47483	CIRCULAR	3	M47510	CIRCULAR	7	M57634	CIRCULAR	4.5
M47484	CIRCULAR	7	M47511	CIRCULAR	0.666667	M57635	CIRCULAR	1
M47485	CIRCULAR	7	M47512	CIRCULAR	0.666667	M57640	CIRCULAR	1.5
M47486	CIRCULAR	2.5	M47513	CIRCULAR	0.666667	M57641	CIRCULAR	3
M47487	CIRCULAR	2.5	M49932	CIRCULAR	7.5	M63251	CIRCULAR	1
M47493	CIRCULAR	2.5	M49933	CIRCULAR	7.5	M63253	CIRCULAR	7
M47494	CIRCULAR	7	M49934	CIRCULAR	9	M73039	CIRCULAR	7
M47500	CIRCULAR	1.5	M49935	CIRCULAR	7	M73727	CIRCULAR	1
M47501	CIRCULAR	1.5	M50012	CIRCULAR	0.666667	M87140	CIRCULAR	2
M47502	CIRCULAR	1.5	M50013	CIRCULAR	0.666667	M87141	CIRCULAR	2.5
M47503	CIRCULAR	0.666667	M50014	CIRCULAR	0.666667	M87837	CIRCULAR	0.833333
M47504	CIRCULAR	0.666667	M50015	CIRCULAR	5.5	M87838	CIRCULAR	1
M47505	CIRCULAR	0.666667	M50022	CIRCULAR	7	M87839	CIRCULAR	5.5
M47506	CIRCULAR	7	M57630	CIRCULAR	4.5	M87840	CIRCULAR	5.5
M47507	CIRCULAR	0.666667	M57631	CIRCULAR	1	M90341	CIRCULAR	0.833333

Appendix E: Stormwater Facility Comparison

Facility Type (ROW only)	About	Cost	Maintenance	Environmental Impacts	Equity
Permeable Pavement (SWMM 3-13)  (via Stormwater Partners)	Pervious pavement allows for water to infiltrate into the ground. Common applications include sidewalks, low-trafficked roadways, and parking lots.	asphalt: $7-$13/sq ft concrete: $8-$16/sq ft	vacuuming at least 2x/year; minimal salt for de-icing; continuous monitoring/inspection	Reduces urban heat, stormwater runoff volume and pollution	(+) cooling (-) high maintenance; risk of clogging and disfunction
Filter Strip (via Minnesota Stormwater Manual)	Gently sloped, narrow vegetated areas	$0.02-$0.10 per sq ft	mowing, weeding, sediment removal, inspection	Reduces urban heat, stormwater runoff volume and pollution	(+) cooling, (+) greenspace (+) aesthetics
Green Street (4-4) (via Green Infrastructure Digest)	Stormwater curb extensions, street planters, rain gardens, swales	depends on facility	DEP is responsible for the maintenance of Green Streets, depends on facility	Reduces urban heat, stormwater runoff volume and pollution Potential for habitat creation	(+) cooling (+) green space (+)aesthetics (+)pedestrian benefits (-)expensive (-)loss of parking
Rain gardens/ bioretention (via Earthwatch)	A shallow depression planted with native plants, underdrain pipe connected for flooding; Collects, stores and allows rainwater from roadway to absorb into ground	depends on site (size, plant selection and density, removal of existing pavement, rerouting downspouts); Avg. installation costs: $4-$35/sqft	Watering weekly during dry season, weeding (by hand), removal of sediment and debris, checking gutters, layer compost or mulch yearly, thin and prune plants as necessary, leaf removal in fall	see above	(+) aesthetics (+) cooling (+) green space (-) expensive
Tree Filters (via Iowa Stormwater)	Mini bioretention boxes filled with a soil mixture, a mulch layer, under-drain system and a shrub or tree Typically used when space is limited; Requires proximity to a storm drain system [8]	~$10-18k	annual check-in, raking at least 2x/yr, replace media every 5-10 years, remove trash and debris	See above + shade	see above
Curb Extensions (via City of Grand Rapids)	Vegetative or sediment catching area between the curb and the sidewalk, Sometimes extended into the parking lane/ pavement is removed to widen the available footprint and treatable drainage area - Requires adequate space and minimal impact to parking	$2,000 to $20,000 per corner (depending on design and site conditions)	n/a	Simple to direct gutter flow into, especially when sited at the downstream end of the flow path.[20]	improve auto, pedestrian, and bicycle safety, improving sight lines, and providing a visual presence in the roadway to slow approaching cars
Grass Swales (biofilters)  (	Allow runoff to percolate into the ground, reducing the amount of runoff leaving the roadway The vegetation covering the side slopes and channel bottom provides a filtration surface for the water and helps to reduce the flow velocity [8] Swales are linear practices, well- suited for treating stormwater from highways or residential roads [16]	$3,500/BMP [14]	cost is 5%-7% of construction	Swales often treat adjacent roadway discharge, which can be a significant source of metals from wear and tear of vehicle components like tires and brake pads [16]	allow for more vegetation, (not always suited for urban areas due to need for larger swales

Sump (4.2.2) (via GSM Plumbing)	"designed to remove rising water from low-lying areas in a building before it can cause damage"[2]	apx. $400/lin ft [13]	clog sump drainage holes and by holding floatable at the surface with a downturned elbow (decreases maintenance needs/increases long-term effectiveness)	achieves pollution reduction benefits needed for UIC	
MSTT	The list of approved MSTTs is posted on the BES website	Depends on facility	installation of sedimentation maintenance hole to provide pretreatment upstream of MSTT	Depends on facility	depends
Filterra Bioretention System (via Contech)	bioretention optimized for high volume/flow treatment and high pollutant removal	depends on size	removal of trash, sediment, mulch; replace mulch layer yearly	same as green streets	same as green streets
Modular Wetlands® Linear Stormwater Bioretention  (via Contech)	stormwater bioretention system to utilize patented horizontal flow (curb inlet, vault, or downspout)	depends on size	remove trash from screening device, remove sediment from separation chamber, replace cartridge filter regularly, replace drain-down media, trim vegetation as needed	high pollutant removal, potential for greenery	can have greenery
PerkFilter® Media Filtration Catch Basin (via Oldcastle)	The PerkFilter® is a media-filled, cartridge filtration system that removes pollutants such as Total Suspended Solids (TSS), metals, nutrients, etc. Can be catch basin, drop inlet, manhole, vault or panel vault	depends on size/configuration; ballpark $500 for precast catch basin [12]	inspection 1-2x/yr, removal of gross pollutants from inlet and treatment chambers, replacement of filter cartridges (depends on site)	high pollutant removal	n/a
Structured Detention (4.2.7) (via Stormtrap)	tanks, vaults, and oversized pipes	$3-10/cf of storage [15]	summer: remove sediment, oil and debris, make necessary structural repairs; winter: monitor flow rate; spring: inspect for structural concerns and sediment accumulation; year-round: clean gutters and rain drains as needed	cannot meet pollution reduction requirements alone	n/a

Appendix F: GSI Function Visualization

Figure A1. Visualization of stormwater planter function (via Philadelphia Water Department)

Figure A2. Visualization of street tree planter function

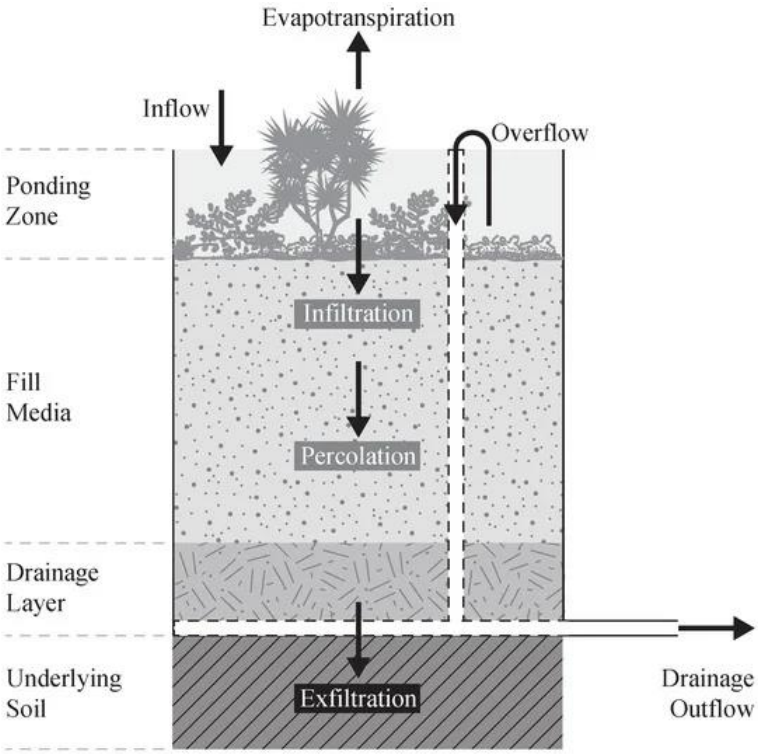
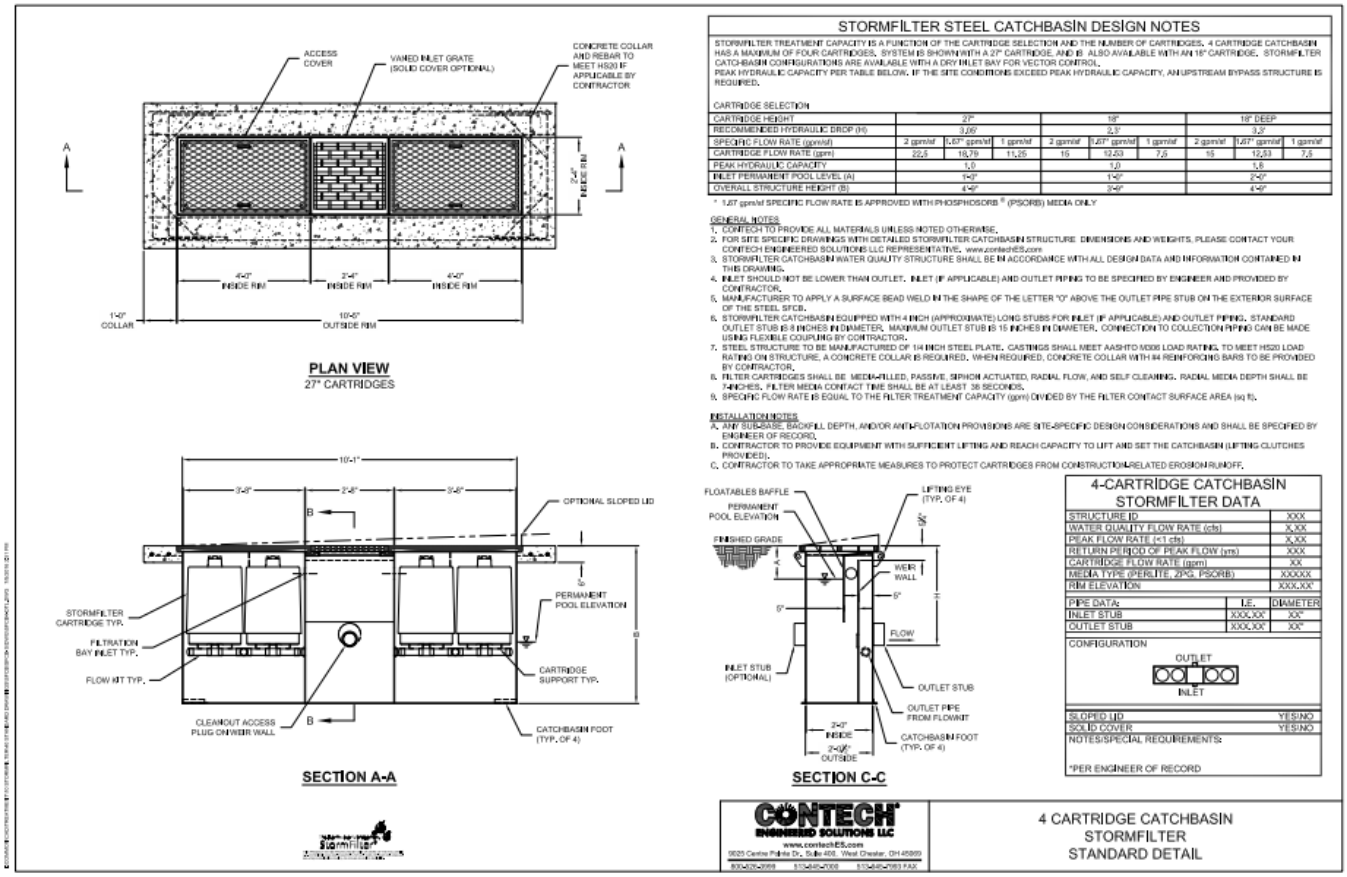


Figure A3. Schematic of a bioretention cell

Appendix G: Stormwater Facility Drawings



STORMFILTER STEEL CATCHBASIN DESIGN NOTES

STORMFILTER TREATMENT CAPACITY IS A FUNCTION OF THE CARTRIDGE SELECTION AND THE NUMBER OF CARTRIDGES. 4 CARTRIDGE CATCHBASIN HAS A MAXIMUM OF FOUR CARTRIDGES. SYSTEM IS SHOWN WITH A 27" CARTRIDGE, AND IS ALSO AVAILABLE WITH AN 18" CARTRIDGE. STORMFILTER CATCHBASIN CONFIGURATIONS ARE AVAILABLE WITH A DRY INLET BAY FOR VECTOR CONTROL.
PEAK HYDRAULIC CAPACITY PER TABLE BELOW. IF THE SITE CONDITIONS EXCEED PEAK HYDRAULIC CAPACITY, AN UPSTREAM BYPASS STRUCTURE IS REQUIRED.

CARTRIDGE SELECTION

CARTRIDGE HEIGHT	27"			18"			18" DEEP		
RECOMMENDED HYDRAULIC DROP (H)	3.05'			2.3'			3.2'		
SPECIFIC FLOW RATE (gpm/sf)	2 gpm/sf	1.67* gpm/sf	1 gpm/sf	2 gpm/sf	1.67* gpm/sf	1 gpm/sf	2 gpm/sf	1.67* gpm/sf	1 gpm/sf
CARTRIDGE FLOW RATE (gpm)	22.5	18.79	11.25	15	12.53	7.5	15	12.53	7.5
PEAK HYDRAULIC CAPACITY	1.0			1.0			1.6		
INLET PERMANENT POOL LEVEL (A)	1'-0"			1'-0"			2'-0"		
OVERALL STRUCTURE HEIGHT (B)	4'-0"			3'-6"			4'-6"		

* 1.67 gpm/sf SPECIFIC FLOW RATE IS APPROVED WITH PHOSPHOSORB ® (PSORB) MEDIA ONLY

GENERAL NOTES
1. CONTECH TO PROVIDE ALL MATERIALS UNLESS NOTED OTHERWISE.
2. FOR SITE SPECIFIC DRAWINGS WITH DETAILED STORMFILTER CATCHBASIN STRUCTURE DIMENSIONS AND WEIGHTS, PLEASE CONTACT YOUR CONTECH ENGINEERED SOLUTIONS LLC REPRESENTATIVE. www.contechES.com
3. STORMFILTER CATCHBASIN WATER QUALITY STRUCTURE SHALL BE IN ACCORDANCE WITH ALL DESIGN DATA AND INFORMATION CONTAINED IN THIS DRAWING.
4. INLET SHOULD NOT BE LOWER THAN OUTLET. INLET (IF APPLICABLE) AND OUTLET PIPING TO BE SPECIFIED BY ENGINEER AND PROVIDED BY CONTRACTOR.
5. MANUFACTURER TO APPLY A SURFACE BEAD WELD IN THE SHAPE OF THE LETTER "O" ABOVE THE OUTLET PIPE STUB ON THE EXTERIOR SURFACE OF THE STEEL SPCB.
6. STORMFILTER CATCHBASIN EQUIPPED WITH 4 INCH (APPROXIMATE) LONG STUBS FOR INLET (IF APPLICABLE) AND OUTLET PIPING. STANDARD OUTLET STUB IS 8 INCHES IN DIAMETER. MAXIMUM OUTLET STUB IS 15 INCHES IN DIAMETER. CONNECTION TO COLLECTION PIPING CAN BE MADE USING FLEXIBLE COUPLING BY CONTRACTOR.
7. STEEL STRUCTURE TO BE MANUFACTURED OF 1/4 INCH STEEL PLATE. CASTINGS SHALL MEET AASHTO M306 LOAD RATING. TO MEET HS20 LOAD RATING ON STRUCTURE, A CONCRETE COLLAR IS REQUIRED. WHEN REQUIRED, CONCRETE COLLAR WITH #4 REINFORCING BARS TO BE PROVIDED BY CONTRACTOR.
8. FILTER CARTRIDGES SHALL BE MEDIA-FILLED, PASSIVE, SIPHON ACTUATED, RADIAL FLOW, AND SELF CLEANING. RADIAL MEDIA DEPTH SHALL BE 7-INCHES. FILTER MEDIA CONTACT TIME SHALL BE AT LEAST 38 SECONDS.
9. SPECIFIC FLOW RATE IS EQUAL TO THE FILTER TREATMENT CAPACITY (gpm) DIVIDED BY THE FILTER CONTACT SURFACE AREA (sq ft).

INSTALLATION NOTES
A. ANY SUB-BASE, BACKFILL DEPTH, AND/OR ANTI-FLOTATION PROVISIONS ARE SITE-SPECIFIC DESIGN CONSIDERATIONS AND SHALL BE SPECIFIED BY ENGINEER OF RECORD.
B. CONTRACTOR TO PROVIDE EQUIPMENT WITH SUFFICIENT LIFTING AND REACH CAPACITY TO LIFT AND SET THE CATCHBASIN (LIFTING CLUTCHES PROVIDED).
C. CONTRACTOR TO TAKE APPROPRIATE MEASURES TO PROTECT CARTRIDGES FROM CONSTRUCTION-RELATED EROSION RUNOFF.

4-CARTRIDGE CATCHBASIN STORMFILTER DATA

STRUCTURE ID	XXX
WATER QUALITY FLOW RATE (cfs)	X.XX
PEAK FLOW RATE (<1 cfs)	X.XX
RETURN PERIOD OF PEAK FLOW (yrs)	XXX
CARTRIDGE FLOW RATE (gpm)	XX
MEDIA TYPE (PERLITE, ZPG, PSORB)	XXXXX
RIM ELEVATION	XXX.XX'

PIPE DATA:	I.E.	DIAMETER
INLET STUB	XXX.XX'	XX"
OUTLET STUB	XXX.XX'	XX"

CONFIGURATION

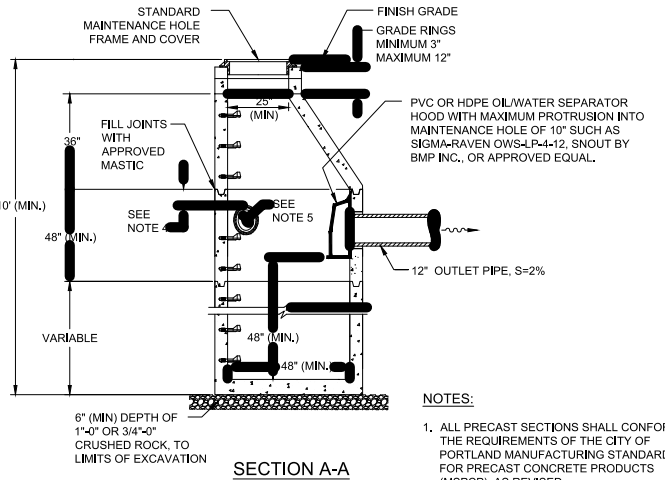
OUTLET

INLET

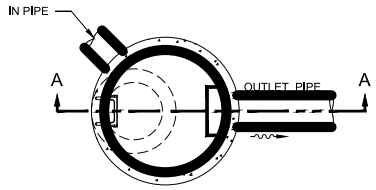
SLOPED LID	YES/NO
SOLID COVER	YES/NO

NOTES/SPECIAL REQUIREMENTS:

*PER ENGINEER OF RECORD



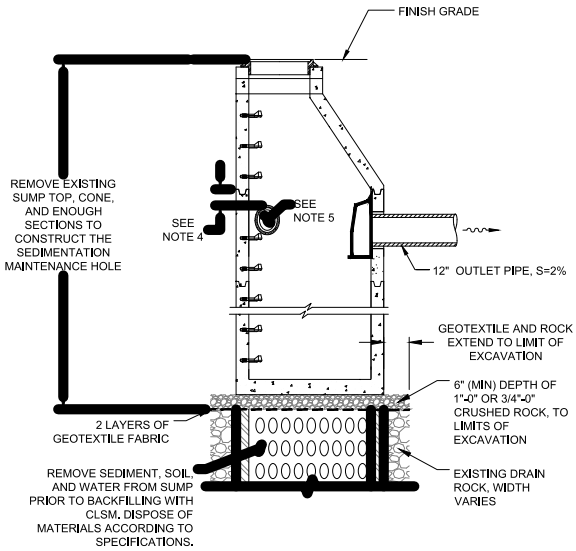
SECTION A-A



PLAN

NOTES:

1. ALL PRECAST SECTIONS SHALL CONFORM TO THE REQUIREMENTS OF THE CITY OF PORTLAND MANUFACTURING STANDARDS FOR PRECAST CONCRETE PRODUCTS (MSPCP), AS REVISED.
2. MAINTENANCE HOLE STEPS – REFER TO THE STANDARD DETAIL P-168.
3. PROVIDE A FLEXIBLE JOINT FOR ALL CONNECTED PIPES:
   - RIGID PIPE < 36 INCHES – 18 INCHES (MAX.) FROM OUTSIDE WALL
   - FLEXIBLE PIPE – 18 INCHES (MAX.) FROM THE OUTSIDE WALL UNLESS A FLEXIBLE JOINT FITTING IS INSTALLED AND ACCEPTED.
4. PROVIDE 6 INCHES (MIN.) OF SEPARATION BETWEEN A SECTION JOINT AND THE OUTER EDGE OF ANY OPENING.
5. LOCATE INVERT OF ANY PIPE 6 INCHES (MIN.) ABOVE THE OUTLET PIPE INVERT.
6. CONTACT THE CSA PROGRAM BEFORE CONVERTING AN EXISTING SUMP TO A SEDIMENTATION MAINTENANCE HOLE.



IN LOCATION OF EXISTING SUMP

The selection and use of this Standard Detail, while designed in accordance with generally accepted engineering principles and practices, is the sole responsibility of the user.

Bureau of Environmental Services
CITY OF PORTLAND, OREGON
[redacted]
Chief Engineer

Standard Detail Title
Sedimentation Maintenance Hole and Sedimentation Maintenance Hole in Location of Existing Sump

Note:
All material and workmanship shall be in accordance with the City of Portland Standard Construction Specifications.

Effective Date: 04-01-23
Calc. Book No.: N/A
Baseline Report Date: N/A

Standard Detail No.
P-161

P-161_SEDIMENTATION_MAINTENANCE_HOLE.DWG 3/21/23 9:52 AM DPERDE

Appendix H: Maps

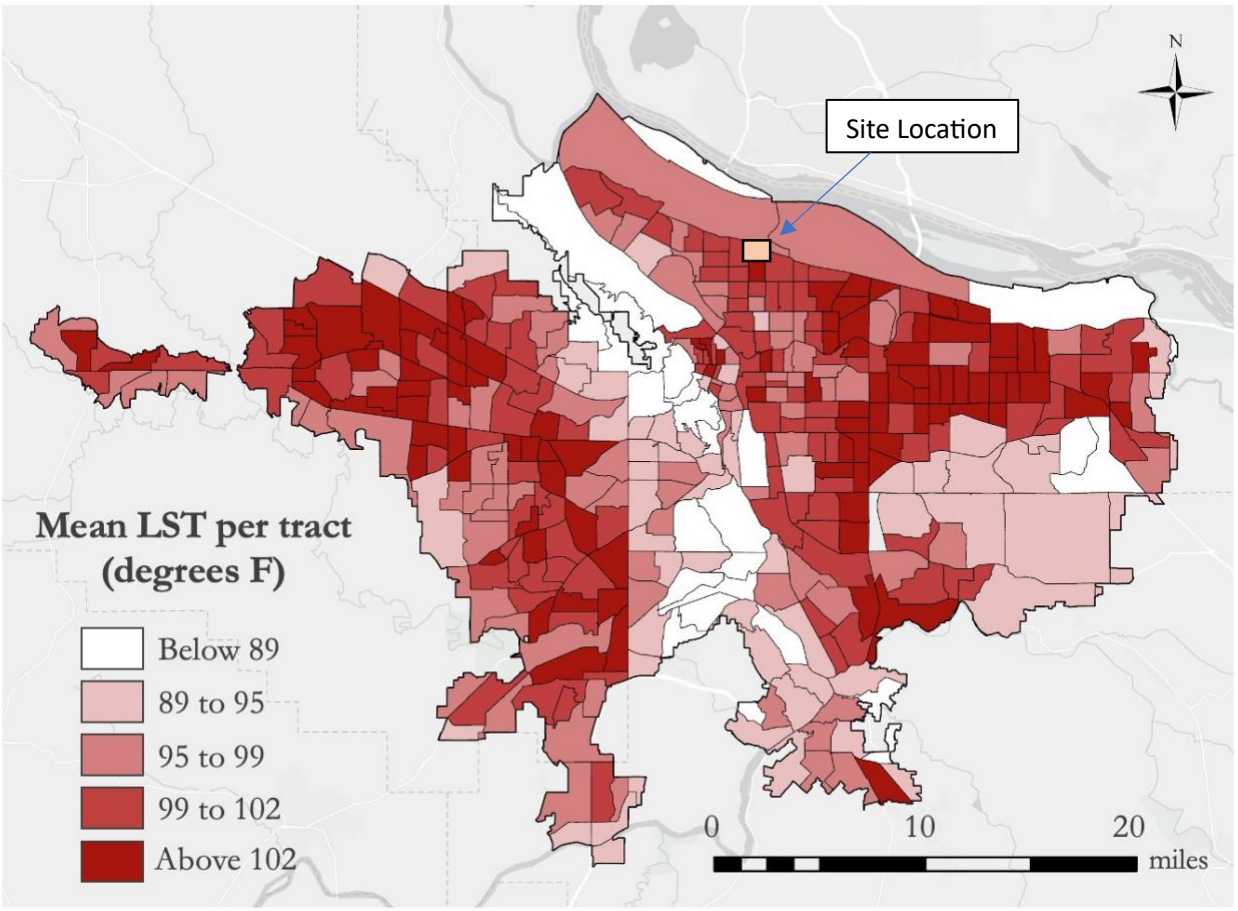


Figure A4. Mean land surface temperature per tract (Source: City of Portland)

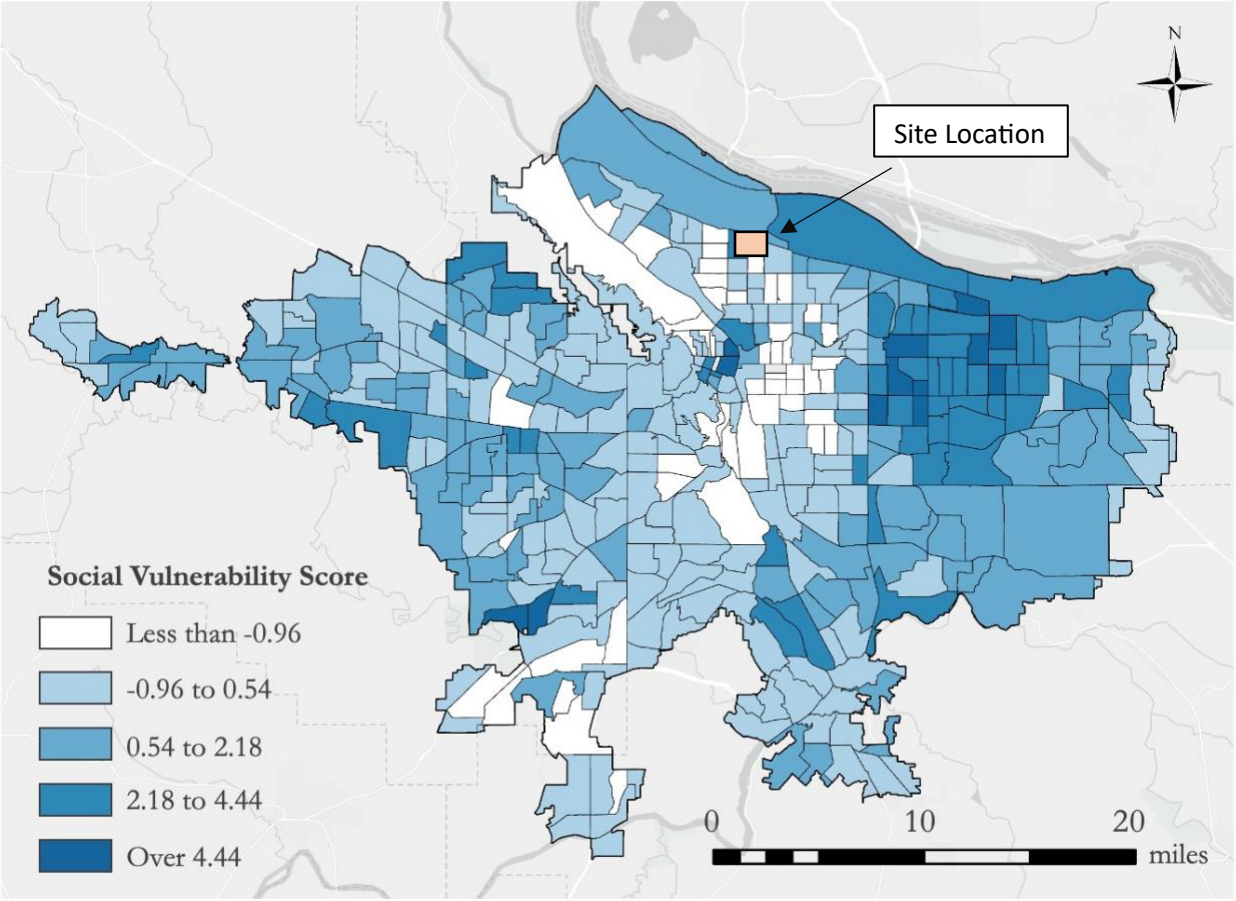


Figure A5. Social Vulnerability Score of Portland Area (Source: City of Portland)